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LARGE LANGUAGE MODELS (LLM): EXAMINING THE QUALITY OF GENERATED TEXT WITH TASK SPECIFIC DATA

BACHELOR THESIS

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Abstract

Large Language Models (LLMs) have emerged as powerful tools for generating human-like text across various applications. This thesis examines the factors influencing the success of LLMs in text generation tasks, focusing on prompting techniques and the integration of external data within the prompt. The research aims to validate the role of prompt engineering in generating compelling content, particularly in the context of writing marketing newsletters.

The thesis begins by introducing the research area of LLMs and prompting, followed by a literary review of relevant prompt engineering techniques. Based on the findings, a set of hypotheses is formulated, outlining the potential impact of these techniques on text generation quality.

A controlled experiment is conducted to validate the hypotheses, applying the identified prompting techniques to the task of writing email newsletters with a single LLM (GPT-3.5-turbo-1106). The generated newsletters are evaluated using quantitative metrics and A/B/n test campaign to an audience (n=550) of registered attendees of a university event.

The results of the experiment confirmed the important role of prompting techniques and external data. On the specific task of newsletter writing, usage of advanced prompting techniques resulted in 15% improvement in click-through rate for a given prompting technique leveraging external data (Improving the click-through rate from 4% to 19%), demonstrating prompt engineering and external data play an important role when creating effective texts that engage the audience.

Techniques leveraging specification of tone of voice, providing additional context data, relevant examples or industry best practice instructions have shown to have the largest impact on the resulting text and achieved superior performance on the measured task compared to a base prompt solution without prompting techniques.

Keywords

AI, Generative AI, context data, LLM, model evaluation, prompt engineering, text generation

Abstrakt

Velké jazykové modely (LLM) se etablovaly jako výkonné nástroje pro generování textů podobajících se lidské řeči v různých aplikacích. Tato práce zkoumá faktory ovlivňující úspěch LLM při úlohách generování textu, s důrazem na techniky psaní úvodních pokynů (promptů) a integraci externích dat do těchto pokynů. Výzkum si klade za cíl ověřit roli systematické tvorby pokynů, známé jako Prompt Engineering, při generování poutavého obsahu, zejména v kontextu psaní marketingových newsletterů.

Práce začíná úvodem do oblasti výzkumu LLM a psaní úvodních pokynů, následovaným literárním přehledem relevantních technik tvorby pokynů. Na základě těchto poznatků je formulována sada hypotéz, které popisují potenciální dopad těchto technik na kvalitu generovaného textu.

K ověření hypotéz je proveden kontrolovaný experiment, při kterém se identifikované techniky psaní pokynů aplikují na úlohu psaní e-mailových newsletterů pomocí jednoho zvoleného velkého jazykového modelu (GPT-3.5-turbo-1106). Generované newslettery jsou hodnoceny pomocí kvantitativních metrik a A/B/n testu emailové kampaně provedené na vzorku (n = 550) registrovaných účastníků univerzitní akce.

Výsledky experimentu potvrdily významnou roli technik psaní pokynů a externích dat. Konkrétně při psaní email newsletterů vedlo použití pokročilých technik psaní pokynů ke 15% zlepšení míry prokliku pro konkrétní techniku využívající externí data (zlepšení míry prokliku ze 4 % na 19 %). To dokládá, že tvorba pokynů a externí data hrají důležitou roli při vytváření efektivních textů, které zaujmou čtenáře.

Techniky využívající specifikaci tónu hlasu, poskytování dodatečných kontextových dat, relevantních příkladů nebo pokynů osvědčených v oboru prokázaly největší vliv na výsledný text a dosáhly lepších výsledků při měřeném úkolu ve srovnání se základním řešením bez pokročilých technik psaní pokynů.

Klíčová slova

AI, Generativní AI, kontextová data, LLM, vyhodnocování modelu, evaluace modelu, prompt engineering, prompt inženýrství, generování textu

Contents

Introduction

Large Language Models (LLMs) have emerged as powerful tools for generating human-like text across various applications. The ability of these models to understand and generate natural language has opened up new possibilities in fields such as content creation, chatbots, or text summarization. However, the quality of the generated text heavily depends on the prompts provided to the model and the techniques used to guide its output.(Kaddour et al., 2023; Webson and Pavlick, 2022; Lu et al., 2022)

This thesis examines the factors influencing the success of LLMs in text generation tasks, focusing on prompting techniques and the integration of external data within the prompt. The research aims to validate the role of prompt engineering in generating compelling content, particularly in the context of writing marketing newsletters.

The first section provides an overview of the research area, highlighting the significance of LLMs and their applications in text generation. It discusses the challenges associated with generating high-quality text and the need for effective prompting strategies. The first sections also outlines the main research questions addressed in the thesis:

Goals and Research Questions

The goal of the thesis is: Evaluate the performance of LLM prompting techniques for a specific text generation task and the integration of external data within the prompt.

The Research questions are:

1. How does the context data used in the prompt relate to the quality of the generated output text?

2. How do different prompt engineering techniques relate to the quality of the output text?

To address these questions, the thesis begins with a literature review of relevant prompt engineering techniques. The review identifies and summarizes prompting techniques applicable to the text generation use case of LLMs. Based on the findings, a set of hypotheses is formulated, outlining the potential impact of these techniques on text generation quality.

Hypotheses: will be defined in chapter 2.3 Hypothesis Formulation

A controlled experiment is conducted to validate the hypotheses, applying the identified prompting techniques to the task of writing invitation newsletters with a single LLM (GPT-3.5-turbo-1106). The generated newsletters are evaluated using quantitative metrics and an A/B/n test campaign to an audience (n=550) of registered attendees of a university event.

Methodology

In this thesis, I plan to employ the following scientific methods:

• Literature review: A comprehensive review of relevant academic papers and research to identify and summarize prompting techniques applicable to the text generation use case of Large Language Models (LLMs).

• Hypothesis formulation: Based on the findings from the literature review, a set of hypotheses will be developed, outlining the potential impact of prompting techniques and the inclusion of external context data on the quality of generated text.

• Computational NLP Analysis: Using techniques such as ROGUE score and Semantic Similarity, I analyse generated text to determine the potential impact of prompting techniques and the inclusion of external context data on the quality of generated text.

• Exploratory Data Analysis: I perform analysis over data generated by my experiment, including analysis of the Computational NLP Metrics and result of my email campaign experiment.

• Controlled experiment: To validate the hypotheses, a multi-step experiment will be conducted, applying the identified prompting techniques to a specific task - email text writing for a real-world conference. The experiment will involve generating email content using a base prompt and various prompting techniques, with the aim of maximizing open rate (OR) and click-through rate (CTR) metrics.

•Abstraction and Concretization: I will employ abstraction and concretization as a pair of research methods, analysing the individual prompting techniques and attempt to understand their impact on text generation based on their theoretical description present in the literature.

•Deduction and Induction: I will utilize deductive reasoning to derive specific hypotheses from the general principles and theories established in the literature review. Conversely, I will employ inductive reasoning to formulate broader generalizations and theoretical contributions based on the empirical findings from the controlled experiment and exploratory data analysis.

Data Collection Methods:

• Recipient dataset: The generated emails will be sent to a dataset of 550 respondents, described in the section 3.4 Dataset Description.

• Email performance metrics: Open rate (OR) and click-through rate (CTR) data will be collected for each email variant to assess the impact of the applied prompting techniques.

Data Analysis and Evaluation Techniques:

• ROUGE-L difference comparison: Comparing the results of each prompting technique to the base prompt using the ROUGE-L metric to assess the degree of difference in the generated text.

• Semantic similarity calculation: Using a transformer model and cosine similarity to compare the results of each prompting technique to the base prompt and a human-written variant created by a domain expert.

• Human evaluation: An expert panel will provide feedback on a shortlist of techniques determined by the ROUGE-L and semantic similarity analyses.

• Statistical analysis: Examining the email performance metrics (OR and CTR) to determine the effectiveness of the applied prompting techniques in achieving the desired outcomes.

1 Description of the Research Area

This research focuses on the area of Large Language Models (LLMs), in the first section, I will give a general overview of the research area of LLMs, Prompt Engineering and text generation use-case.

1.1 Large Language Models

LLMs have emerged as a result in scaling large (most commonly) transformer-based neural networks. (Devlin et al., 2019) demonstrated scaling such a model from previously used thousands of parameters to numbers counting millions or more parameters in a single model. This has been shown to enable language models to generalize to a large variety of text-based tasks with no or minimal task-specific architecture modifications. Additionally scaling the training dataset from smaller curated datasets to very large and broad datasets, spanning large chunks of the public internet enabled further scaling of such models. (Brown et al., 2020)

Comparison	**Traditional ML**	**Deep Learning**	**LLMs**
Training Data Size	Large	Large	Very large
Feature Engineering	Manual	Automatic	Automatic
Model Complexity	Limited	Complex	Very Complex
Interpretability	Good	Poor	Poorer
Performance	Moderate	High	Highest
Hardware Requirements	Low	High	Very High

Figure 1 - Taxonomy of ML and LLMs (Chang et al., 2023)

Task and domain generalization, enabled by scaling of text-based models has shifted the paradigm of "train-finetune-and-predict" of previous models to an interaction based on providing text-based instructions and examples, known as prompts, to guide the model output. This process of dynamically adapting to tasks based on prompts is called in-context learning (Min et al., 2022)

1.1.1 Prompt

Prompt can be defined as a "*natural language input, into a language model instructing it to generate a certain output*." (Marvin et al., 2024)

'Language Models are Few-Shot Learners', the original paper introducing GPT-3, describes prompting as a zero-shot capability of language models, enabled by generalization from its pre-training dataset on a diverse set of tasks (Brown et al., 2020). NLP models have historically been demonstrated to benefit from being trained on multi-task datasets, therefore being able to generalize among multiple tasks with distinct outputs (Caruana, 1997). This involved mixing multiple input-output datasets into a single supervised training process.

Today, prompts include various types of information including instructions, questions and other details such as context, inputs, or examples.

Since prompts play a critical role in properly leveraging the capabilities of LLMs, a lot of research has appeared attempting to systematically evaluate and design prompts. From this practice, a domain called prompt engineering has emerged.

I will describe the role of **prompt engineering** and **prompting techniques** in section 1.4.4. But first, I will give an overview of the use cases of LLMs and their evaluation.

1.2 Common Use Cases of Large Language Models

Large language models (LLMs) have a wide range of applications across various domains and tasks due to their ability to understand and generate human-like text. Initially, language models started as a text completion tool, able to finish pre-written sentences (Radford et al., 2019). With better models and fine-tuning approaches, novel LLMs have much broader capabilities. Additionally, with increasing adoptions, its users have found novel ways to leverage LLMs for a wide range of specific tasks.

Here, I explore some of the most common use cases of LLMs.

1.2.1 Text Generation

Text generation is the simplest use case demonstrating the current capabilities of LLMs, ranging from simple text completions of older models to modern chatbot-style LLMs capable of generating diverse types of content such as:

- stories
- marketing texts and content
- articles
- emails
- essays

and many others.

In “Language Models are Few-Shot Learners” the original paper introducing GPT-3, the authors demonstrate a use case of generating news articles, by instructing the model to create new articles based on providing a few examples of articles from a news website (therefore using the few-shot approach). In their experiment, unlike previous (smaller) models, human participants generally could not distinguish the generated article from real articles.

(Brown et al., 2020)

Text generation is the main focus of the thesis, since Large Language Models are currently employed across various fields to generate a wide range of content. To more clearly delineate the specific use case of text generation, I will contrast it with description of other distinct use cases of LLMs, including code generation, classification, and question answering.

1.2.2 Questions Answering

LLMs have been trained as next-word predictors, effectively modelling language including its semantic complexity. But along with the increasing scale of LLMs an ability to retain information on variety of domains, present in its training data, has enabled the use of LLMs for question answering.

‘Language Models as Knowledge Bases?’(Petroni et al., 2019), shows the potential use case for language models in open-domain question answering. The authors investigate the ability of pretrained language models like BERT to answer questions without any fine-tuning or access to a dedicated information retrieval system.

They test this by manually converting a subset of questions from the SQuAD dataset (Stanford Question Answering Dataset, a text comprehension dataset) into cloze-style queries. They find that the large BERT model achieves impresive results on this open-domain question answering task, reaching 57.1% precision, nearly matching a dedicated open-domain question answering system (DrQA) that uses a supervised relation extraction component and information retrieval (scoring 63.5%) (Petroni et al., 2019).

The authors note that this "surprisingly strong ability of these models to recall factual knowledge without any fine-tuning demonstrates their potential as unsupervised open-domain QA systems" (Petroni et al., 2019). They suggest that large pre-trained language models like BERT could potentially serve as viable alternatives or complements to traditional knowledge bases extracted from text for open-domain question answering and other tasks that require accessing factual knowledge.

Newer models have demonstrated even stronger abilities beating state-of-the-art QA systems. GPT-3 has historically shown strong zero-shot, one-shot, and few-shot performance on several open-domain QA datasets like TriviaQA and Natural Questions. On TriviaQA, GPT-3 outperforms a fine-tuned 11B parameter model by over 14% in the zero-shot setting. With few-shot examples, GPT-3 matches the state-of-the-art performance of a learned retrieval system over a 21M document corpus (Brown et al., 2020).

Despite the impressive performance of large language models like GPT-3 on many question-answering benchmarks, it is important to note that these models have also demonstrated a tendency to "hallucinate" incorrect answers with high confidence when faced with inputs outside their training distribution. This remains an unresolved challenge that limits the trustworthiness and safe deployment of such models for open-domain QA applications or other use-cases (Ji et al., 2023).

The issue of hallucinations and generating unsupported outputs is further explored in subsequent sections of this thesis. Potential mitigation strategies are also discussed.

1.2.3 Language Translation

The GPT-3 paper also explored the ability of large language models to perform machine translation in an unsupervised or few-shot learning setting. In contrast to typical unsupervised neural machine translation (NMT) approaches that pretrain on parallel monolingual datasets and use backtranslation, GPT-3's pretraining data contained a mix of non-English text blended together with the primarily English data (Brown et al., 2020).

An interesting finding was that few-shot GPT-3 significantly outperformed previous unsupervised approaches when translating into English, showcasing its strength as a highly capable English language model. However, it underperformed when translating out of English into other languages like French, German, and Romanian (Brown et al., 2020).

For some higher-resource language pairs like French-English and German-English, few-shot GPT-3 even exceeded the performance of supervised NMT models from the literature, though the authors note these may not represent true state-of-the-art supervised systems. For lower-resource Romanian-English, GPT-3 came close to matching a strong supervised system that used 608K labeled examples and backtranslation (Brown et al., 2020; Jiao et al., 2023).

Newer and more powerful models have been released that surpass the capabilities of GPT-3 for machine translation. One major breakthrough was the introduction of GPT-4 by OpenAI in March 2023. Jiao et al. (2023) provide an in-depth evaluation of using GPT-4 as the translation engine.

Their study found that GPT-4 boosts translation performance compared to GPT-3 significantly across multiple language pairs, bringing it to the level of top commercial translation systems like Google Translate. On benchmarks like WMT for translating between distant language pairs like Chinese-English, GPT-4 achieved competitive or even superior performance compared to previous state-of-the-art systems (Jiao et al., 2023).

1.2.4 Classifications

Large language models have shown impressive performance on various classification tasks, even with limited or no task-specific fine-tuning. This includes sentiment analysis, topic classification, and intent detection, among others.

For example, the GPT-3 paper (Brown et al., 2020) demonstrated strong results on several classification benchmarks in the zero-shot and few-shot settings. On the SST-2 sentiment analysis dataset, GPT-3 achieved 80.2% accuracy with zero-shot learning and 92.5% accuracy with one-shot learning, approaching the performance of state-of-the-art fine-tuned models. Similarly, on the AG News topic classification dataset, GPT-3 reached 65.6% accuracy with zero-shot learning and 77.6% accuracy with one-shot learning.

1.2.5 Code Generation

Large language models have also demonstrated remarkable capabilities in generating code, making them valuable tools for software development and programming assistance. By learning from vast amounts of publicly available code, these models can understand and generate code in various programming languages, assisting with tasks such as auto-completion, bug fixing, and even writing entire functions or programs from natural language descriptions. (Kojima et al., 2023; OpenAI et al., 2024; Chen et al., 2023a; Feng et al., 2023)

1.2.6 Agentic Behaviour

Another trend connected to the rapidly improving performance of LLMs is use of the term LLM Agent. Despite historical mentions of the term agent varying context across machine learning, LLM Agents demonstrated by (Yao et al., 2023) introduced a framework called ReAct where the LLM is used to generate both: reasoning steps and task-specific actions in an alternating manner.

ReAct can be viewed as a combination of different prompting techniques described in the later section including Chain-of-thought (CoT) prompting and Multi-step prompting. Similarly, ReAct has shown task-specific performance increase (Yao et al., 2023).

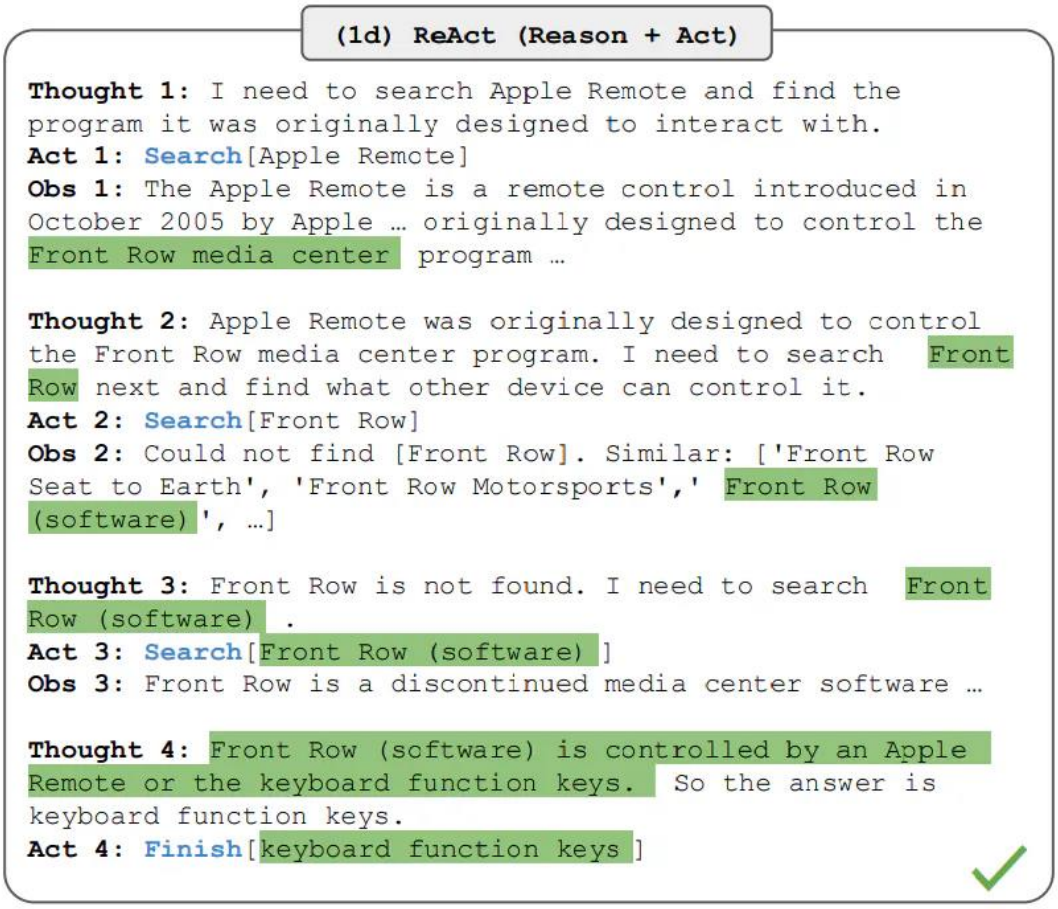

Figure 2 - ReAct demonstration (Yao et al., 2023)

Additionally, ReAct enables another phenomenon connected with LLM agents: tool use. By instructing the model to generate a step including a use of an external tool and subsequently consume the tool's output, the LLM Agent can make use of various external tools including web-search, calculator, python shell and other (Paranjape et al., 2023).

Multi-agent conversation workflows:

LLMs have also been explored for their potential in multi-agent conversation workflows, where multiple LLMs collaborate and coordinate to achieve shared goals or solve complex problems. (Wu et al., 2023a) demonstrated that LLMs can engage in multi-agent conversations, where they can take on different roles, exchange information, and coordinate their actions to complete tasks that require multiple perspectives or specialized knowledge.

1.3 Evaluation of Large Language Models

This section will explore various approaches and methodologies for evaluating LLMs, highlighting the key differences from traditional machine learning evaluation techniques. I present a common taxonomy of LLM evaluation methods, covering human evaluation, natural language processing (NLP) evaluation, LLM-based evaluation, and programmatic evaluation techniques.

1.3.1 Traditional Machine Learning Evaluation

In traditional machine learning, models are typically evaluated using a variety of metrics such as accuracy, precision, recall or F1-score. These metrics are well-established and widely used for tasks such as classification, regression, and clustering. The evaluation process commonly involves splitting the data into training, validation, and test sets, while using the test set to assess the model's performance on unseen data (Raschka, 2020).

1.3.2 LLM Evaluation and its Key Differences

Evaluating large language models (LLMs) presents unique challenges due to their ability to perform a wide range of tasks, including text generation, question answering, and reasoning. Unlike traditional machine learning models, LLMs are not typically trained on a specific task or dataset, but rather on vast amounts of text data from the internet. As a result, the evaluation process for LLMs requires a different approach.

Raschka (2020) frames the key difference between traditional LLM evaluation where we ask “Where” (datasets, benchmarks) and “How” to evaluate ML models, LLMs pose the additional dimension of “What” to evaluate. Furthermore, topics such as evaluation of robustness, ethics, biases, and trustworthiness remain a large area of research interest.

Some of the most common evaluation goals are:

- Accuracy - a measure of how correct a model is on a given task.
    - Common metrics: F1 score, Confusion Matrix (precision, accuracy, ..), ROGUE score
- Robustness - performance of a model in the face of various challenging inputs
    - Common metrics: Attack Success Rate (ASR), Performance Drop Rate (PDR), various benchmarks for testing variability of the input
- Alignment with human preference - assesses the degree to which the language model’s output aligns with human values, preferences, and expectations
    - Common metrics: bias score, fairness assessments. eg: Demographic Parity Difference (DPD), Equalized Odds Difference (EOD)

(Raschka, 2020)

Based on (Raschka, 2020) we can further divide the taxonomy of LLM evaluation, focusing on the “Where” question into:

- General LLM benchmarks - aimed at evaluating the multitude of capabilities of LLMs, therefore serving best for LLM model comparison and selection
- Benchmarks for Specific Downstream Tasks - aimed at evaluating LLMs for a specific, narrow task, therefore best serving for evaluation of a certain LLM in a specific setting allowing for optimization or finetuning.

1.3.3 General LLM Benchmarks

Several benchmarks and leaderboards have been developed to evaluate the performance of LLMs across various tasks and domains. These include:

- MMLU (Massive Multitask Language Understanding): The Multitask Mutated Language Understanding benchmark is a collection of tasks designed to test language understanding capabilities, including question answering, coreference resolution, and natural language inference.
- BIG-bench: BIG-bench is a collaborative benchmark intended to test LLMs on a diverse set of tasks contributed by the research community, covering areas such as logical reasoning, common sense reasoning, mathematics, code comprehension, and more.(Feng et al., 2023)
- HELM (Holistic Evaluation of Language Models): HELM is a benchmark for tasks such as sentiment analysis, named entity recognition, and reading comprehension. It aims to provide a comprehensive assessment of a model's language understanding abilities.
- SuperGLUE: SuperGLUE is a benchmark for evaluating general-purpose language understanding systems. It consists of a series of challenging natural language understanding tasks, such as question answering, textual entailment, and coreference resolution.
- TruthfulQA: TruthfulQA is a benchmark that evaluates a model's ability to generate truthful and informative responses to questions. It tests the model's factual knowledge, reasoning capabilities, and ability to provide accurate and relevant answers.

(Srivastava et al., 2023; Hendrycks et al., 2021; Raschka, 2020; Chang et al., 2023)

1.3.4 Evaluating LLMs on Specific Downstream Tasks

While general LLM benchmarks provide a broad assessment of model capabilities across various tasks, evaluating LLMs on specific downstream tasks allows for a more targeted and in-depth analysis of their performance in particular domains or applications. This approach enables researchers and practitioners to optimize and fine-tune LLMs for specific use cases, ensuring their effectiveness and suitability for real-world deployment.

Task-specific evaluation typically involves curating datasets, designing metrics, and establishing baselines that are tailored to the unique requirements and challenges of the downstream task at hand. By focusing on a narrower scope, task-specific evaluation can provide more actionable insights and facilitate the development of specialized LLM solutions.

Some common approaches to task-specific LLM evaluation include:

1. Domain-specific benchmarks: Creating benchmark datasets and tasks that are representative of the target domain, such as legal, medical, or financial text processing. These benchmarks allow for a more accurate assessment of LLM performance in the context of domain-specific language, terminology, and reasoning requirements.
2. Application-oriented metrics: Designing evaluation metrics that align with the specific goals and success criteria of the downstream application. For example, in a text summarization task, metrics such as ROUGE (Recall-Oriented Understudy for Gisting

Evaluation) or BERTScore can be used to measure the quality and relevance of generated summaries against human-written references.

3. Human evaluation: Engaging domain experts or end-users to assess the quality, usability, and appropriateness of LLM-generated outputs for the specific task. Human evaluation can provide valuable qualitative insights that may not be captured by automated metrics alone, such as the coherence, fluency, and contextual relevance of the generated text.

(Chang et al., 2023)

1.3.5 Evaluation of Text Generation Tasks

Evaluating the performance of language models on text generation tasks, such as open-ended dialogue, creative writing, and summarization, presents unique challenges due to the absence of a reference answer (often referred to as a "golden" answer). Unlike many other natural language processing tasks, where the goal is to match a predetermined ground truth, text generation often involves producing novel, creative, and contextually appropriate content (Chang et al., 2023; Celikyilmaz, Clark and Gao, 2021)

Here, I present the two, most common, approaches for open-ended text generation evaluation, reference-based evaluation and reference-free evaluation (Chang et al., 2023; Celikyilmaz, Clark and Gao, 2021).

Reference-based Evaluation:

One approach to evaluating text generation is to compare the model's output to an existing (most often human-written) reference text. This method, often referred to as reference-based or supervised evaluation, relies on the availability of high-quality reference data. Common metrics used in this approach include:

NLP-based metrics: ROUGE-1, ROUGE-2, and ROUGE-L measure the overlap between the generated text and reference texts at different n-gram levels.

Semantic similarity: Metrics like BERTScore and METEOR assess the semantic similarity between the generated text and reference texts, capturing meaning beyond surface-level word overlaps.

While reference-based evaluation provides a quantitative measure of similarity to human-written texts, it has limitations. It may fail to capture the diversity and creativity of language models, as they can generate high-quality outputs that deviate from the reference data.

(Celikyilmaz, Clark and Gao, 2021)

Reference-free Evaluation:

To address the limitations of reference-based evaluation, reference-free or unsupervised evaluation methods have been developed. These methods aim to assess the quality and diversity of the generated text without relying on reference data. Some examples include:

Perplexity: This metric measures the probability of the generated text according to the language model itself, providing an estimate of its fluency and coherence.

Self-BLEU: By computing the BLEU score between different segments of the generated text, self-BLEU can measure the diversity and novelty of the output.

Mauve: This learned metric, trained on human judgments, aims to capture various aspects of generated text quality, such as coherence, fluency, and specificity.

LLM-as-a-judge: LLMs themselves can be used as judges or evaluators for certain tasks, such as classification, verification, and fact-checking. In this approach, the LLM is fine-tuned or prompted to assess the outputs or predictions of other LLMs or models, based on its own understanding of the task and the available data.

(Chen et al., 2023b; Celikyilmaz, Clark and Gao, 2021; Raschka, 2020; Zheng et al., 2023)

While reference-free evaluation methods can provide valuable insights into the quality and diversity of generated text, they may not directly assess the factual accuracy, relevance, or appropriateness of the output for a specific task or context.

Given the inherent complexity of text generation tasks and the absence of a single ground truth, a combination of reference-based and reference-free evaluation methods, along with human evaluation, offers the best option to gain a comprehensive understanding of a language model's performance and capabilities.

1.4 Optimization of Large Language Models for a Specific Task

Multiple methods are currently widely used to optimise LLMs on specific tasks, ranging from specific prompting strategies to methods diving deeper into the mechanistic principles inside LLMs. In this section I will briefly describe 4 different strategies for optimizing the performance of an LLM for a specific task or domain, concluding with prompt engineering, the main focus of my research.

1.4.1 LLM Sampling Parameter Tunning

LLM Sampling Parameter tuning (also sometimes referred to as: API Parameter Tuning, Hyperparameter Tuning, Configuration Parameter Tuning) is the process of adjusting the settings of an LLM to adjust its behaviour and possibly optimize its performance for a specific task. There are 2 of such parameters most commonly seen implemented in current LLMs including GPT-3.5, and GPT-4 (Brown et al., 2020; OpenAI et al., 2024):

- Temperature: Controls the randomness of the generated output. Higher values lead to more diverse outputs, while lower values result in more deterministic outputs.
- Top-P and Top-K: These parameters limit the pool of possible next words based on their probability distribution. Top-P selects the smallest set of words whose cumulative probability exceeds a specified threshold, while Top-K selects the K most likely next words.

1.4.2 Fine-tuning

Fine-tuning involves training an LLM on a specific dataset to adapt it to a particular domain or task. This approach can have a high impact but is more useful in specific scenarios and requires more data. There are two main types of fine-tuning:

- Supervised fine-tuning: Training the LLM on a labelled dataset to improve its performance on a specific task, such as sentiment analysis or named entity recognition. (Balaguer et al., 2024)
- Parameter-Efficient Fine-Tuning (PEFT): A technique that fine-tunes only a small portion of the model's parameters, reducing the computational cost and data requirements compared to full fine-tuning. (Balaguer et al., 2024)

Additionally, Reinforcement Learning and Reinforcement Learning with Human Feedback (RL and RLHF) (OpenAI et al., 2024; Brown et al., 2020) are other approaches to fine-tuning LLMs. These methods involve training the model through interaction with an environment or human feedback, allowing it to learn from rewards and penalties.

1.4.3 Custom Model Development

Developing task-specific custom LLMs from scratch is a possible approach to tailoring models for specific domains or tasks. For example, Bloomberg's BloombergGPT (Wu et al., 2023b) demonstrated a custom, 50 billion parameter language model, trained on financial data. Training of BloombergGPT involved a proprietary 345 billion token dataset and extensive investment into training. This model has shown superior performance on domain-specific benchmarks, beating at the time state-of-the-art LLMs. Despite this effort, BloombergGPT has been surpassed by newer generalist models including the latest version of GPT-4 on many benchmarks. This, including the extensive effort and investment by Bloomberg team, shows the immense difficulty in building custom LLMs.

We can conclude that custom model development might not be feasible for most individuals due to the significant computational resources and data required. The foundational papers for LLMs, such as GPT-2 and GPT-3, highlight the scale and complexity involved in training these models from scratch (Radford et al., 2019; Brown et al., 2020).

1.4.4 Prompt Engineering

Referring to our previously used definition of a prompt as a “natural language query, directing the LLM's output” a process of systematically designing and optimizing is commonly referred to as “**Prompt Engineering**”.

The quality of the prompt can have a drastical impact on the performance of the LLM on a downstream task (Kaddour et al., 2023; Webson and Pavlick, 2022; Lu et al., 2022). Despite that, the optimal methods for engineering such prompts is currently an open question. A broad range of prompt engineering techniques has emerged, each showcasing a potential performance improvement on a specified set of tasks.

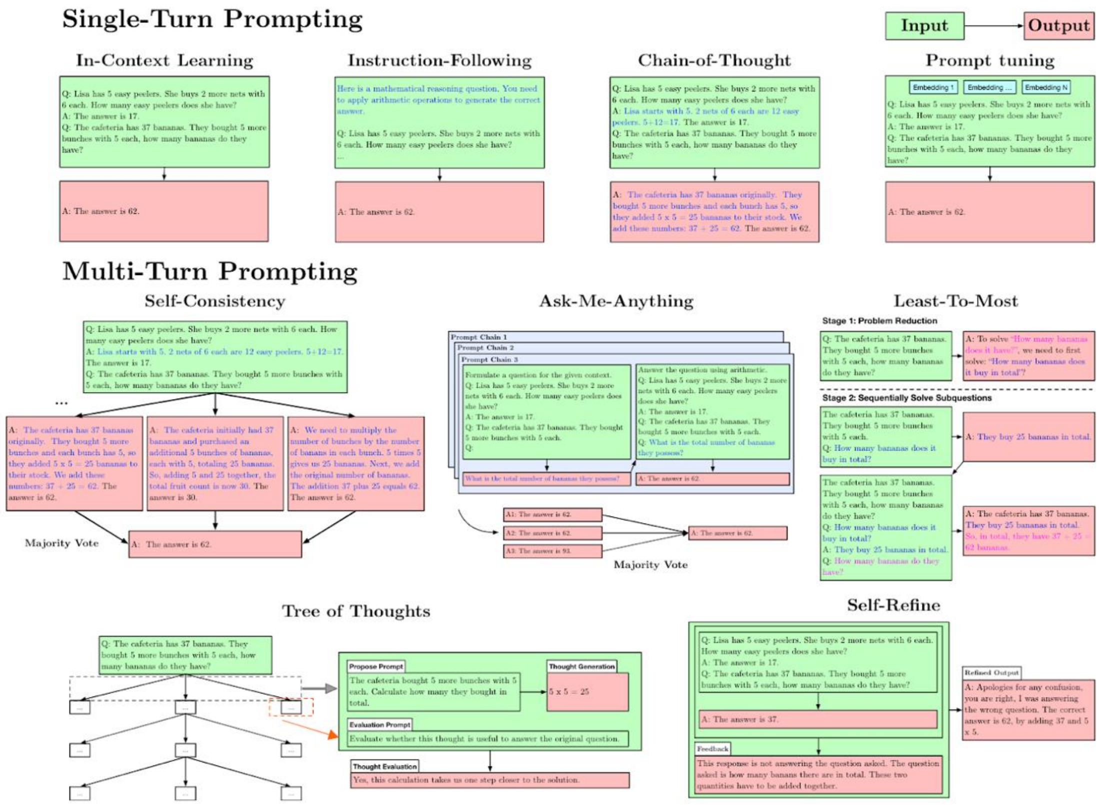

Figure 3 - Prompting techniques overview (Kaddour et al., 2023)

These techniques consist of a set of practices that can be used to refine an initial prompt (base-prompt). For example, Chain-of-Thought is a technique consisting in instructing the model to generate intermediate reasoning steps. Its implementation consists in including: either an example of intermediate reasoning or an instruction to first generate suchreasoning, before giving the final answer. This prompt engineering technique has been shown to improve given LLMs performance on a multitude of benchmarks, as shown in figure 4 (Wang et al., 2023a; Wei et al., 2023; Kojima et al., 2023)

	Arithmetic					
	SingleEq	AddSub	MultiArith	GSM8K	AQUA	SVAMP
zero-shot	74.6/**78.7**	**72.2/77.0**	17.7/22.7	10.4/12.5	22.4/22.4	58.8/58.7
zero-shot-cot	**78.0/78.7**	69.6/74.7	**78.7/79.3**	**40.7/40.5**	**33.5/31.9**	**62.1/63.7**
	Common Sense		Other Reasoning Tasks		Symbolic Reasoning	
	Common SenseQA	Strategy QA	Date Understand	Shuffled Objects	Last Letter (4 words)	Coin Flip (4 times)
zero-shot	**68.8/72.6**	12.7/**54.3**	49.3/33.6	31.3/29.7	0.2/-	12.8/53.8
zero-shot-cot	64.6/64.0	**54.8**/52.3	**67.5/61.8**	**52.4/52.9**	**57.6/-**	**91.4/87.8**

Figure 4 - Zero-shot-CoT evaluation results (Kojima et al., 2023)

Importantly, authors of this paper acknowledge the fact that this technique has shown a neutral or even negative impact on performance on certain benchmarks. This illustrates an important fact about prompt engineering: The usage of prompt engineering techniques leads to a potential improvement only for certain tasks and use-cases.

2 Overview of Prompt Engineering Techniques Relevant for Text Generation

In this section, I will perform a literary review aimed at identifying prompt engineering techniques relevant for the use case of text generation. I conduct this review with the main goal to find sources relevant for the two main research questions standing at the core of this research:

1. How does the context data used in the Prompt relate to the quality of the generated output text
2. How do different Prompt Engineering techniques relate to the quality of the output text

2.1 Literature Review

To identify relevant literature for the research questions, a systematic search was conducted using the following search terms:

(llm || large language model) + (prompting || prompt engineering) ? techniques

(llm || large language model) + instruction

The search was performed on two primary databases: Google Scholar and ArXiv. These databases were chosen for their comprehensive coverage of academic research and their focus on computer science and artificial intelligence.

In addition to the systematic search, the method of snowballing was employed, starting with key academic papers in the subject area. This included seminal works such as the paper introducing the Attention Mechanism (Vaswani et al., 2023) and the paper describing GPT-3 (Brown et al., 2020). By examining the references and citations of these foundational papers, additional relevant literature was identified, ensuring a comprehensive coverage of the research topic.

Inclusion criteria were applied to ensure the relevance and quality of the selected papers. Papers were included if they presented findings relevant to the research questions. Manual filtering was performed to assess the relevance of each paper to the specific focus of the thesis. Preference was given to newer sources to capture the most recent developments in the rapidly evolving field of LLMs and prompt engineering. Additionally, papers with a higher number of citations were prioritized, as they are more likely to have had a significant impact on the research community.

Exclusion criteria were also employed to maintain the scope and feasibility of the research. Papers not written in English or Czech were excluded, as these are the languages accessible to the researcher. Furthermore, papers for which full access could not be obtained were excluded, as a comprehensive review of the content was necessary for accurate analysis and interpretation.

By applying these search terms, databases, inclusion criteria, exclusion criteria, and the snowballing method, a focused and relevant set of literature was identified to inform the research questions and guide the development of hypotheses for the experimental phase of the thesis.

2.2 List of Prompting Techniques Relevant for Text Generation

After performing literary research, I've distilled individual prompt engineering techniques relevant to text generation into a set of the following techniques. Each technique describes a specific approach to constructing prompts used with LLMs, that promises a downstream task improvement.

I Introduce a set of 10 unique techniques, with their description, variants, and possible implementation.

Existing literature on prompt engineering generally contains mentions of prompting techniques, often under various synonyms, including: “Prompting principles”, “Prompting Methods”, “Prompting Tricks”, “Prompting Tactics”, “Prompting Strategy”. For the purpose of this research, I will use the term “Prompting Technique”.

One area with limited research is the transferability of such techniques across different LLM models and their versions. Most of the research focuses on the use of models by OpenAI, GPT-3, GPT-3.5, and GPT-4. Despite that, most papers including this thesis attempt to generalize findings on a single model into techniques generally applicable to other models and their versions with limited effort on localization.

The prompting techniques and their corresponding sources generally focus on demonstrating the performance improvement resulting from the use of such techniques on a given benchmark. Perhaps for this reason, the majority of the sources mention describe and evaluate the use on easily measurable use-cases such as classification or question answering.

I further divided these techniques into 2 high-level categories:

A. prompting techniques leveraging existing knowledge and capabilities of the LLM

   Relating to our research question 2.

B. prompting techniques based on providing external information to the prompt of the LLM

   Relating to our research question 1.

A) Prompting Techniques Leveraging Existing Knowledge and Capabilities of the LLM

1. Chain-of-Thought Prompting / Generated Knowledge

Chain-of-thought (CoT) prompting is a technique that encourages the language model to break down complex problems into a series of intermediate reasoning steps before providing the final answer. By prompting the model to generate a chain of thought, it can better leverage its inherent knowledge and problem-solving capabilities to arrive at more accurate and well-reasoned responses (Wei et al., 2023; Kojima et al., 2023).

The process of CoT prompting typically involves presenting the model with a prompt that includes a few-shot example of intermediate questions or steps, guiding it towards the final solution. For instance, in an arithmetic problem, the prompt might break down the question into smaller sub-problems, encouraging the model to solve each step and provide a detailed explanation of its reasoning process. By structuring the prompt in this manner, users can effectively steer the model's thought process, resulting in more accurate and transparent outputs.

In implementation, multiple methods aim to leverage this concept. Zero-Shot-CoT introduced in ‘Large Language Models are Zero-Shot Reasoners’ (Kojima et al., 2023) uses the phrase “Let's think step by step” in the initial prompt, instructing the model to solve the problem incrementally and generate intermediate reasoning steps. Meanwhile the original paper introducing (few-shot) CoT uses a demonstrative example (few-shot) including the intermediate reasoning steps as a guide for the final output (Wei et al., 2023).

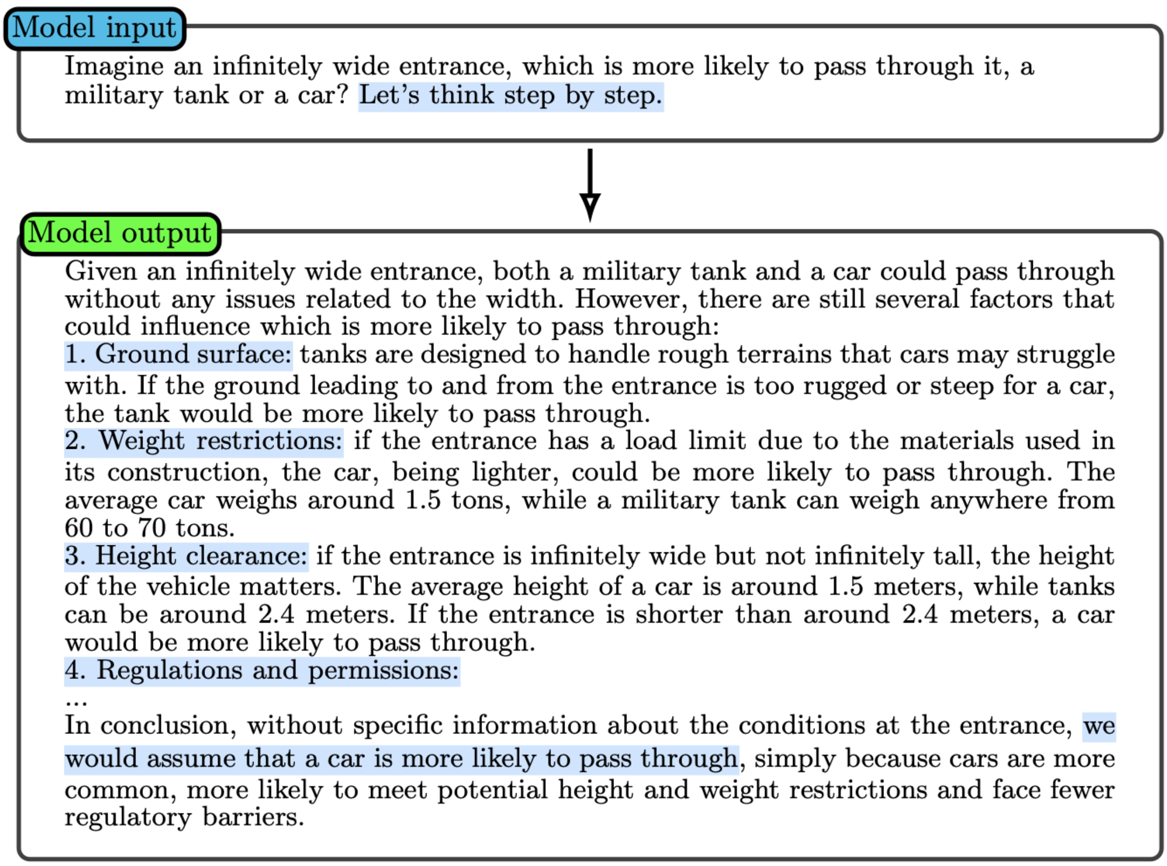

Figure 5 - Zero-shot-CoT Prompting technique demonstration (Chen et al., 2023a)

‘Large Language Models are Zero-Shot Reasoners’ (Kojima et al., 2023) demonstrates a significant leap in specific tasks with specific benchmarks including performance on arithmetic reasoning benchmarks like MultiArith and GSM8K. For instance, with the Zero-shot-CoT method, accuracy on the MultiArith task surged from 17.7% to 78.7%, and on the

GSM8K task, it improved from 10.4% to 40.7% using the InstructGPT model (text-davinci-002). On different tasks including common sense reasoning with SenseQA benchmark, Zero-Shot-CoT has shown no measurable improvement. This highlights the specificity of this technique and the promised improvement to a certain subset of tasks.

Generated knowledge
Another variant of this technique based on a similar concept is “Generated Knowledge Prompting” (Liu et al., 2022).

Similarly, this technique relies on the ability of the LLM to improve its result if it is given more space (tokens) to generate. Main difference form CoT is that Generated Knowledge instructs the model to generate relevant knowledge, not reasoning steps. Specifically, this technique is applied in cases where combining more knowledge is required and therefore might be more suitable for text generation tasks, where success doesn't depend on a single true/false aspect but rather on a larger set of interviewing components.

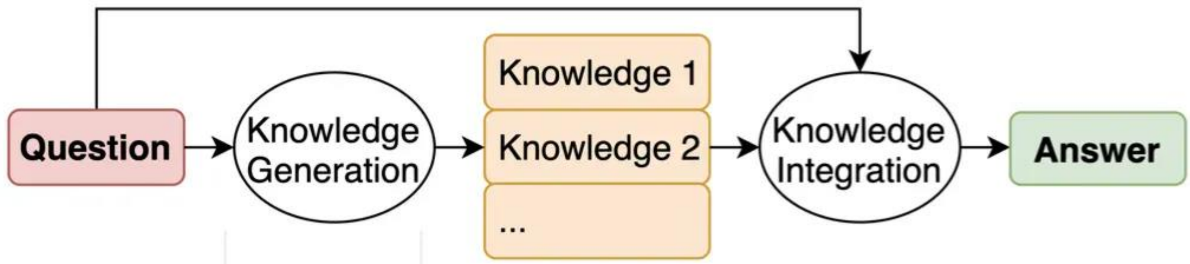

Figure 6 - Generated Knowledge Prompting technique demonstration (Liu et al., 2022)

2. Prompt Structure

Following the definition of a prompt as a “natural language query directing the outputs of an LLM”, use of natural language implies a large degree of freedom, in how we construct the prompt. Despite that, various techniques and findings have emerged linking certain styles of writing prompts as superior to other (Bsharat, Myrzakhan and Shen, 2024; Chen et al., 2023a; White et al., 2023; OpenAI, 2023; Bsharat, Myrzakhan and Shen, 2024).

OpenAI’s official prompt engineering guide suggests the use of delimiters as separator for individual parts of the prompt, that includes use characters such as # (hashtags), “”” (triple quotes) or separators eg. *<article> article text here </article>* Use of these special characters and separators and the ability of the LLM to process them can likely be attributed to their vast present in the training and finetuning data (OpenAI, 2023).

(Bsharat, Myrzakhan and Shen, 2024) suggest a combination of special characters and prompt sections in combination with other prompting techniques including few-shot prompting. Specifically, they suggest use of the following structure:

```
###Instruction### - containing the task description and what is required
by the LLM
###Example###’ - containing examples of how the task could be solved based
on the few-shot approach
```

###Question### - containing the question to the LLM, if relevant

(Bsharat, Myrzakhan and Shen, 2024)

Similarly, viral article outlined a potential framework for combing techniques: CO-STAR (Jules S. Damji, 2024), describes a prompt template consisting in sections conveying context, objective, style, tone, audience and response format.

Example 1

In [8]:

```
user_prompt = """
# CONTEXT #
I want to share our company's new product feature for
serving open source large language models at the lowest cost and lowest
latency. The product feature is Anyscale Endpoints, which serves all Llama series
models and the Mistral series too.

#############

# OBJECTIVE #
Create a LinkedIn post for me, which aims at Gen AI application developers
to click the blog link at the end of the post that explains the features,
a handful of how-to-start guides and tutorials, and how to register to use it,
at no cost.

#############

# STYLE #

Follow the simple writing style common in communications aimed at developers
such as one practised and advocated by Stripe.

Be perusaive yet maintain a neutral tone. Avoid sounding too much like sales or marketing
pitch.

#############

# AUDIENCE #
Tailor the post toward developers seeking to look at an alternative
to closed and expensive LLM models for inference, where transparency,
security, control, and cost are all imperatives for their use cases.

#############

# RESPONSE #
Be concise and succinct in your response yet impactful. Where appropriate, use
appropriate emojies.
"""
```

Figure 7 - Prompt Structure Prompting technique demonstration (Jules S. Damji, 2024)

3. Role Prompting / Roleplay

The Roleplay prompting techniques is based on the fundamental principles behind current LLM. As base models are trained, they go through a sequence of fine-tunning and instruction tunning steps aimed at aligning the model with the instruction / prompt based mode of interaction (Ouyang et al., 2022; Brown et al., 2020). During these steps, base model is trained to follow a conversation style dialogue, consisting of exchanging roles between the LLM and the user. These steps help improve the instruction following of the LLM but also align the models output with human preferences. This inherently results in an emergence of a “persona” representing the role of an LLM. Not only does the resulting LLM convey human preferences and behaviour to a surprisingly good degree (Zheng et al., 2023), it also open up the possibility to alter this persona with prompt engineering and replicate such emergent properties, only tailored to a more specific role.

(White et al., 2023) describes “The Persona Patter” as a technique instructing the model to always take a point-of-view or perspective of a specific persona. Showcased on the role of a security expert with a prompt:

"From now on, act as a security reviewer. Pay close attention to the security details of any code that we look at. Provide outputs that a security reviewer would regarding the code."

(White et al., 2023)

This example was highlighted as a quick method helping constrain and focus the LLM to the topic of a security expert and help align the generated text with this role.

(Shanahan, McDonell and Reynolds, 2023) challenge the antrophormisation of LLMs by showcasing the apparent lack of other aspects of humane behaviour such as self-awareness. Despite that, roleplay prompting technique has remained as a strong option to quickly align the LLM and the likelihood of specific text/language being preferred during the mechanistic next-word prediction process with the goals of the prompt engineer.

4. Self-reflection / Multi-step-prompting

Despite the counter intuitive nature, multiple sources have shown that LLMs are better at evaluating their own responses than at initially producing them (Zheng et al., 2023; Yang et al., 2022; Madaan et al., 2023).

This results in the possibility to use self-reflection as an intermediate step, while producing results. Most commonly implemented in a multiple-step patter, where we first instruct the model to generate a result with a (base) prompt and subsequently ask the model to generate feedback, criticism or review of its own response. Lastly, the LLM is instructed to refine / regenerate its output based on the feedback. This can result in measurable performance improvement, commonly demonstrated on question-answering tasks (Yang et al., 2023, 2022).

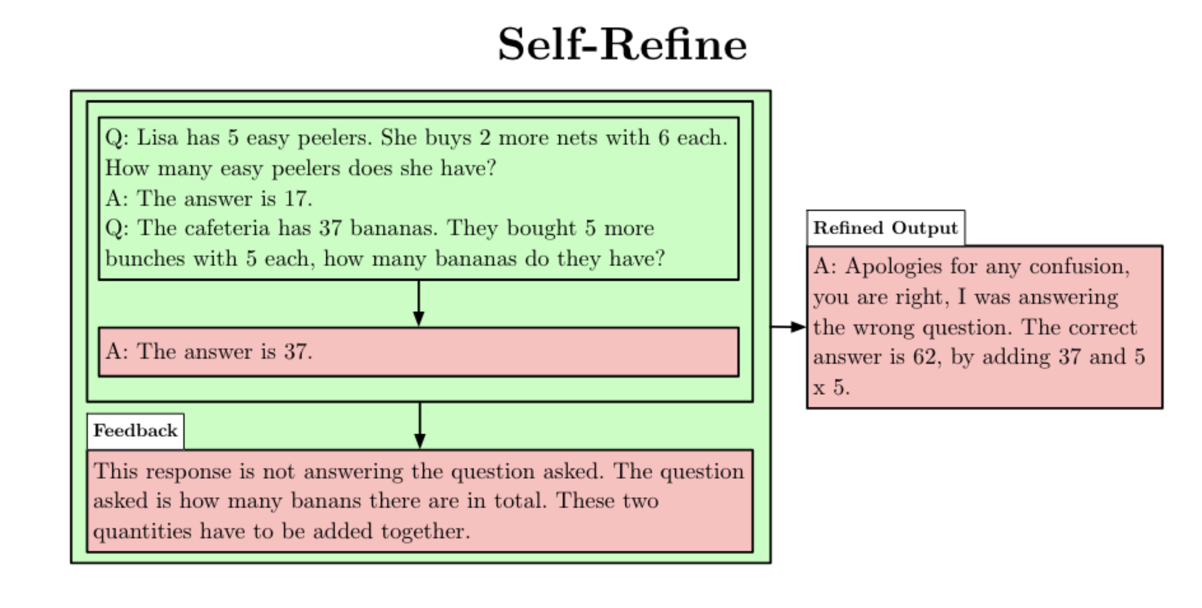

Figure 8 - Self-Refine Prompting technique demonstration (Shanahan, McDonell and Reynolds, 2023)

(Madaan et al., 2023) showcase and evaluate an implementation of such approach. Their results showcase a noticeable improvement among multiple domains and tasks, notably:

1. Dialogue Response Generation: The SELF-REFINE approach led to substantial improvements in generating dialogue responses. Using GPT-4, the base model had a performance score of 25.4%, which increased to 74.6% with SELF-REFINE, showing an absolute improvement of 49.2% in aligning with human preferences and generating more contextually relevant and engaging responses.
2. Code Optimization: In tasks involving code optimization, the application of SELF-REFINE to GPT-4 increased the performance from 27.3% to 36.0%, an improvement of 8.7%. This reflects the model's enhanced ability to generate more efficient and correct code based on iterative feedback and refinements provided by itself.

Yang et al. showcases a more complete implementation of self-reflection combined with multi-step prompting (Yang et al., 2022). The authors propose a sequential set of prompt with reflection cycles (shown in figure 9), designed to improve long-form text generation.

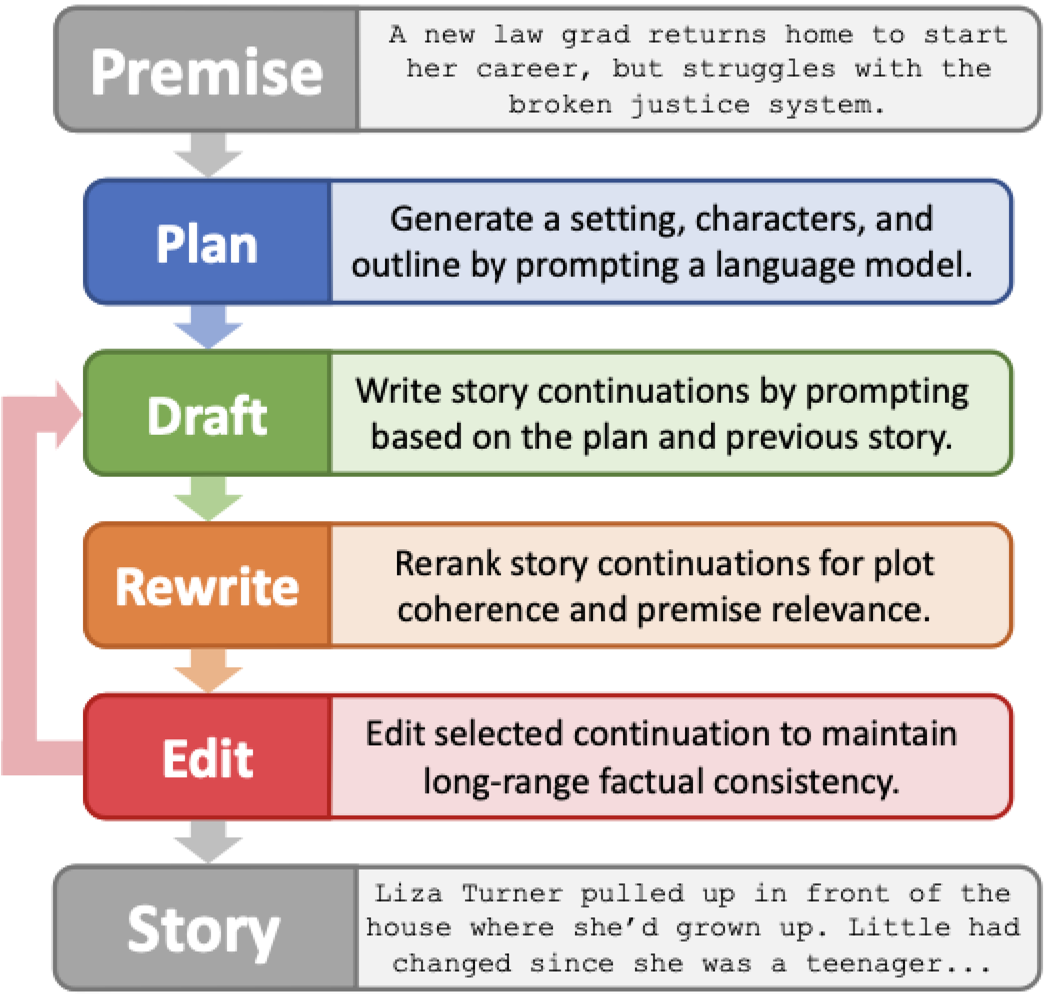

Figure 9 - Re3 (Yang et al., 2022)

"Compared to similar-length stories generated directly from the same base model, human evaluators judged substantially more of Re3's stories as having a coherent overarching plot (by 14% absolute increase), and relevant to the given initial premise (by 20%)." (Yang et al., 2022)

These results show strong promise in extracting better texts with current models with structured methods. In my conclusion, Re3 can better fit into a category of Agentic LLM approaches due to its complexity and combination of different prompting techniques and not as a standalone prompting technique.

5. Self-Consistency

Self-consistency, as introduced by (Wang et al., 2023b) is an approach that involves querying an LLM multiple times with the same prompt and taking the majority result as the final answer. This technique builds upon the foundation of chain-of-thought (CoT) prompting which encourages the model to break down complex problems into intermediate reasoning steps. By combining self-consistency with CoT prompting, researchers have demonstrated significant improvements in the performance of LLMs across a range of tasks.

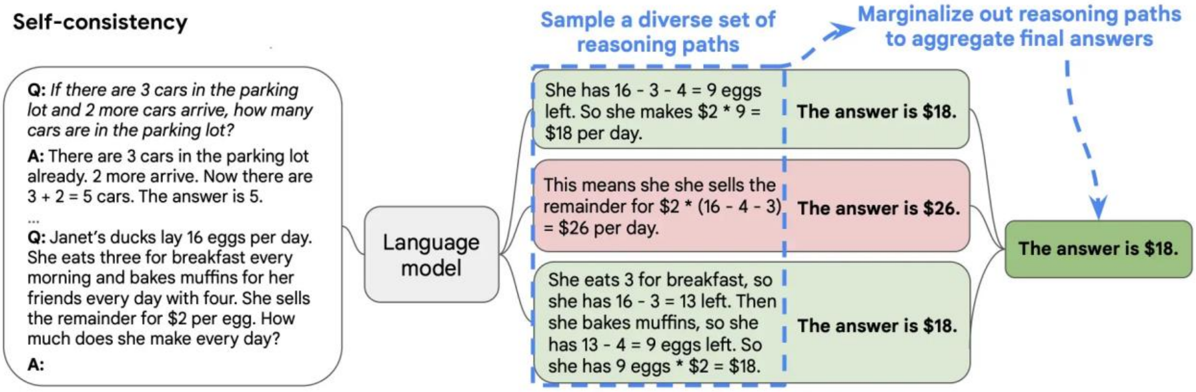

Figure 10 - Self-Consistency Prompting technique demonstration (Wang et al., 2023b)

The effectiveness of self-consistency lies in its ability to mitigate the inherent stochasticity and variability in LLM outputs. By generating multiple independent chains of thought for a given prompt, self-consistency allows for the aggregation of diverse perspectives and reasoning paths. The majority voting mechanism then selects the most commonly occurring answer, which is more likely to be correct and reliable compared to individual outputs.

(Wang et al., 2023b) conducted extensive experiments to evaluate the impact of self-consistency on various benchmarks, including arithmetic reasoning, common-sense reasoning, and symbolic reasoning tasks. Their results showed that self-consistency consistently improved the performance of LLMs, even in cases where regular CoT prompting was found to be ineffective (Wang et al., 2023b). For instance, on the MultiArith dataset self-consistency boosted the accuracy of the GPT-3 model from 17.7% to 78.7%, demonstrating its ability to enhance the model's arithmetic reasoning capabilities.

No work was found demonstrating concrete use of self-consistency for generating longer texts nor its impact on the text quality.

6. Incentives and Keywords

The application of specific incentives has appeared in various forms across the literature. Incentives convey the use of specific phrases supposed to express urgency, importance and significance of the used prompt.

(Bsharat, Myrzakhan and Shen, 2024) use prompt “I’m going to tip $300000 for a better solution!”. In a specific benchmark (designed by the authors of the paper) the use of this prompt has lead to a 40% improvement in precision.

In other case a Twitter user showcases an experiment where using a similar prompt offering a tip to the LLM results in generation of longer responses in open-domain writing task

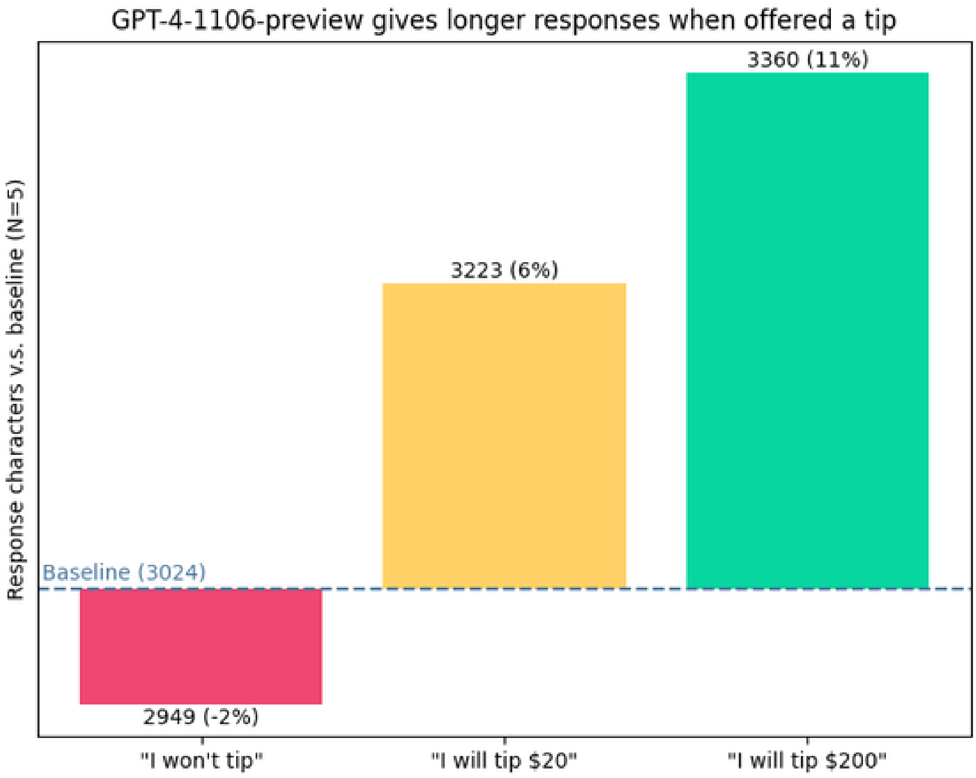

Figure 11 - Incentives evaluation (Vogel, 2023)

In ‘Large Language Models Understand and Can Be Enhanced by Emotional Stimuli‘(Li et al., 2023) authors test LLM behaviour when using emotional stimuli inside the prompt as shown in figure 12. Use of emotional cues referred to as “EmotionPrompt”, they've achieved a measurable improvement in certain benchmarks including 8.00% relative performance improvement in Instruction Induction benchmark and 115% in BIG-Bench benchmark. This paper also shows a noticeable improvement on human evaluation of open-ended text concluding a 10.9% average improvement in terms of performance, truthfulness, and responsibility metrics (Li et al., 2023).

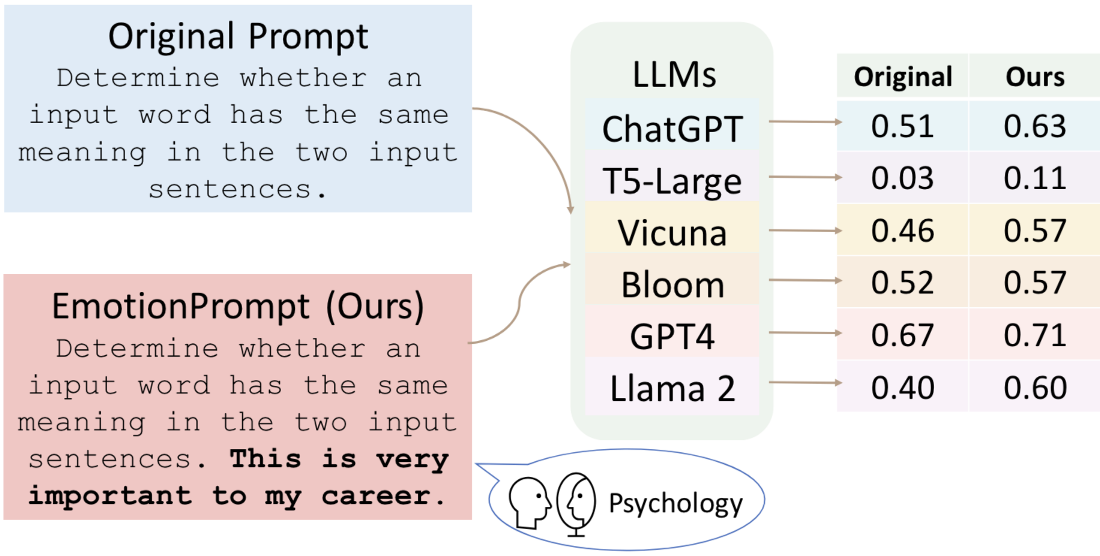

Figure 12 - Emotional Stimuli Prompting technique *(Li et al., 2023)*

(Li et al., 2023) propose a possible explanation of this effect. By demonstrating the impact of using emotional stimuli on a smaller Flan-T5-large language model, results show the model calculate higher attention score for the contents of the prompt when paired with an emotional stimulus. Additionally, the authors argue that use of such stimuli changed and increased the likelihood distribution of certain tokens during generation, possibly resulting in the increase in human preference ranking.

B) Prompting Techniques Based on Providing External Information to the Prompt of the LLM

Models have all their information stored inside their parameters, allowing them to function to some extend as a knowledge base (Petroni et al., 2019). But accessing the information contained inside the LLM can be troublesome, additionally LLMs knowledge has limits and does not contain any proprietary or user-specific data. To address this limitation, number of techniques has emerged. Most common name for this approach is Retrieval Augmented Generation (RAG) (Lewis et al., 2021). This technique consists in querying an external knowledge base for additional information, that is subsequently passed inside the prompt along with the instructions. The domain of RAG is generally divided into 3 distinct sections, corresponding to the sequential steps when applying RAG. Each section has its specifics and knowhow. They are:

- Knowledge storage method
- Retrieval search function
- Prompt Engineering the result prompt

For the purpose of this thesis, we will disregard the first and second sections, as they are large and complex domain within themselves and offer opportunities for different research efforts. For my research, I will focus on the question relevant to the last section. Firstly *What* types of data might be included inside the prompt and secondary *How* to best tie them together with the prompt.

Retrieval Augmented Generation additionally plays a role when addressing hallucinations. Hallucination is an issue of LLMs and refers to the phenomenon where the content generated by the model is nonsensical or unfaithful to the provided source content (Chang et al., 2023). By systematically curating the external data within RAG, users can achieve higher accuracy of results and therefore reduce hallucinations.

7. Including the target audience

This technique consists in providing information about the target audience in the prompt.

(Bsharat, Myrzakhan and Shen, 2024) demonstrate the effectiveness of providing information about the audience of the result. Their example includes the phrase "the audience is an expert in the field." This technique can result in outputs that are more tailored for the specified recipient. A common example of this technique consists in asking for an explanation of a difficult topic.

"When providing external information to the prompt, it is important to consider the target audience. The information should be tailored to the specific needs and background knowledge of the intended users. For example, if the target audience is domain experts, the included information can be more technical and assume a higher level of prior knowledge. On the other hand, if the target audience is the general public, the information should be more accessible and provide necessary context and explanations" (OpenAI, 2023)

8. Specifying the Tone of Voice and Style

As the text generation during inference follows a probabilistic representation of the training data, LLMs output generally result in a very neutral tone. Despite that, LLMs or other transformer-based models including text-to-image generators, have shown an incredible capability at following style and style-specific instructions given in prompts (Liu and Chilton, 2022).

Specification of style and tone has gained a lot of attention with text-to-image generative models. In 'Design Guidelines for Prompt Engineering Text-to-Image Generative Models' (Liu and Chilton, 2022) discuss the possibilities of using specific aspects of the prompt such as "Style keywords" and their impact on the resulting image.

This prompting technique consists in specifying the tone and style of the desired output in the prompt. This technique allows for a broad range of options and creativity on the side of the prompt engineer, but therefore limits the ability to systematically test its efficiency. The requested style and tone can either be described and profiled with instructions in the prompt.

This approach, can be considered a zero-shot equivalent of few-shot learning for style transfer, allows for greater control over the generated text without the need for explicit examples (OpenAI, 2023).

9. Including Relevant Samples

Few-shot learning, a paradigm in which a model learns from a limited number of examples, has been a long-standing approach in machine learning. Despite the advent of large language models (LLMs) and their impressive capabilities, few-shot learning remains relevant and effective in the context of prompt engineering for text generation tasks.

One prominent technique in few-shot learning is providing relevant samples as part of the prompt. By including examples that demonstrate the desired output, the LLM can better understand the task at hand and generate text that aligns with the given samples. This approach is particularly useful when aiming to guide the model towards a specific style, structure, or format.

When providing samples for text generation, the traditional few-shot approach is difficult, since the task has no golden answer or perfect solution. But we can consider the angle of style or structure transfer. By carefully selecting examples that exhibit the desired characteristics, such as a particular writing style, tone, or organizational structure, the LLM can learn to generate text that mimics those attributes. This technique allows for greater control over the generated output and enables the creation of text that adheres to specific guidelines or requirements. (OpenAI, 2023)

This paradigm can be especially useful for writing domain-specific texts. For instance, if the goal is to generate a formal business letter, providing samples of well-written business correspondence can help the LLM understand the expected tone, language, and structure. Similarly, if the aim is to generate creative writing pieces, such as poetry or short stories, including relevant examples can guide the model towards producing text with the desired literary elements and style.

(Li et al., 2018) present a retrieval augmented generation approach based on retrieving similar texts from a set corpus for text generation tasks, with the goal of transferring style and sentiment.

(Krishna, Wieting and Iyyer, 2020) introduced a transformer-based model for style transfer already in 2020 with an encoder style architecture. In their testing, the model showed impressive capabilities, even for single-shot (single sample) style transfer. Later models, including GPT-3 showed impressive style transfer with a decoder style model.

10. Including External Data Relevant to the Task

By providing the LLM with additional context or knowledge beyond the instructions within the prompt itself, the model can generate text that is more accurate, informative, and aligned with the specific requirements of the task as shown by (Lewis et al., 2021; Li et al., 2022).

Including external data in the prompt allows the LLM to leverage information that may not be part of its pre-existing knowledge. This is particularly useful when generating text related to current events, specific industries, or specialized domains. By supplying the model with up-to-date facts, statistics, or domain-specific terminology, the generated text can be more factually accurate and relevant to the intended audience (Lewis et al., 2021).

For example, if the task is to generate a news article about a recent scientific discovery, including key details about the research, such as the names of the scientists involved, the institution where the study was conducted, and a brief summary of the findings, can help the LLM produce a more comprehensive and accurate report. Similarly, if the goal is to generate product descriptions for a specific industry, providing the model with relevant product features, specifications, and benefits can result in more persuasive and informative text.

When including external data in the prompt, it is essential to ensure that the information is reliable, accurate, and relevant to the task. LLM users should carefully curate the data and present it in a clear and concise manner to avoid overwhelming the model with excessive or irrelevant details. Additionally, it is important to consider the format in which the data is presented, such as using structured templates or separating the external information from the main prompt using delimiters. (OpenAI, 2023; Li et al., 2022, 2018; Lewis et al., 2021)

2.3 Hypothesis Formulation

Based on the observed techniques, the literature suggests use of prompting techniques promises improvement on a downstream task. Given the focus of the experiment outlined in the further sections of this thesis, a set of hypotheses may be outlined. These hypotheses attempt to reflect the suggested techniques behind prompting principles should apply in our selected experimental task of text generation.

A "**Base Prompt**" is constructed as a reference solution without using any advanced prompting techniques, that should correspond to a simple LLM user prompt. Further description of the base prompt including its formulation is described in section 3.3. Methodology of the Experiment

H 1. For a defined and specific LLM task, we get significantly better results when applying more advanced prompting techniques than a base prompt solution.

H 2. For a defined LLM text generation task, such as newsletter writing, applying advanced prompting techniques will result in generated text that is more similar to a preferred human-generated solution than a solution using a base prompt.

H 3. In a domain-specific LLM task, such as newsletter writing, incorporating external data, including industry best practices, academic studies, and expert instructions, into the prompts will significantly improve the quality of the generated text compared to prompts without this additional information.

2.3.1 Hypothesis Evaluation Methodology

I plan to evaluate the hypotheses using the following methods:

For H1, H3 I will evaluate the results of a specific text generation task (email newsletter writing) through task-specific evaluation metrics (open rate, click-through rate). I will compare the results to the result of a base prompt solution.

For H2 I will measure the Semantic Similarity measured by cosine distance compared to a human-written variant created by an expert. I will also collect qualitative feedback from an expert panel.

3 Experiment: Examining the Quality of Generated Text with Prompting Techniques and Task Specific Data

The goal of the experiment is to evaluate the hypotheses outlined in the previous section, and investigate the impact of prompting techniques and inclusion of external context data, as per my research questions. The experiment can be considered as quasi-representative because of the sample for the A/B/n test audience.

For this purpose, I've outlined a multi-step controlled experiment consisting of first using intermediary evaluation techniques (ROGUE-L, Cosine distance Semantic Similarity) to perform a Computational Analysis aiming to measure the impact of a wide range of prompting techniques. Secondly, based on the results of the Computational Analysis with ROUGE-L and Cosine Semantic Similarity, I narrow down a shortlist of techniques that will undergo a human evaluation experiment.

The human evaluation experiment will be conducted with 2 methods:

1. Expert panel evaluation (with 2 experts).
2. Email Newsletter Campaign A/B/n test (with an audience of 550 respondents).

The specific task selected to conduct our experiment is **email text writing**, focusing on creating the contents of an email. The goal of the email campaign is to address the audience of a conference realized in March 2024 at VŠE Prague. Particularly, we want to thank the attendees for their presence and provide a link to an article written about the conference. The task was specified in conjunction with a stakeholder group, consisting of the conference organizers.

Our task has the following requirements defined by the respective stakeholders:

Criteria #1. The email must stay on topic of summarizing the conference and referencing the article

Criteria #2. The email must contain the provided link

Criteria #3. The email must not contain untrue or misleading information

3.1 The Role of Text Quality in Email Marketing

Email is a widely used method of Digital Marketing. It establishes a direct channel for brands, businesses, and other parties to directly engage with their audience. It allows for a wide range of use-cases ranging from invitations, offerings, informative newsletters and many others. As email marketing continues to evolve, the question of what constitutes effective email content has been a subject of interest for both businesses and researchers (Conceição and Gama, 2019).

One aspect of content quality is the relevance of the content to the user. This aspect is difficult to control, and irrelevant content has a low chance of effectively grabbing attention of the recipient. However, there are many factors email creators can influence (Conceição and Gama, 2019). The quality of the text plays a crucial role in determining the success of email marketing campaigns. Factors such as subject line, location of URL links, and the length of the invitation text can significantly influence the engagement and conversion rates of emails. Crafting relevant and compelling content is essential to capture the attention of recipients and encourage them to take the desired action (Kaplowitz et al., 2012).

3.2 Measuring the Quality and Performance in Email Marketing

The quality of email in email marketing is generally measured by its performance metrics - KPIs. Among the most used KPIs are open rate and click-through rate. In some use-cases further metrics can be measured such as conversion rate or other criteria relevant to the specific business and its goals.

Open-rate - measures the percentage of recipients who open an email compared to the total number of emails sent

Click-through rate (CTR) - the percentage of recipients who click on one or more links contained in an email compared to the total number of emails sent.

(Kolowich, L., 2016)

To effectively measure and compare performance of emails, controlled experiments can be conducted. Controlled experiment in email marketing measures the performance of different variants of the same email in a controlled environment. This type of test is often called A/B test and the practice is called A/B testing. Since my experiment uses multiple different variants within a single experiment, I will use the commonly seen term A/B/n test. Major requirement for running successful A/B/n tests is to clearly distinguish between the controlled variable and the subject of the test. For example, to test the efficiency of different subject line + body text combination, it should be tested without changing the email layout, topic of the email or its audience (Kohavi et al., 2009).

3.3 Methodology of the Experiment

The order methodology of my experiment is described in this section. The experiment and its individual steps are outlined in figure 13.

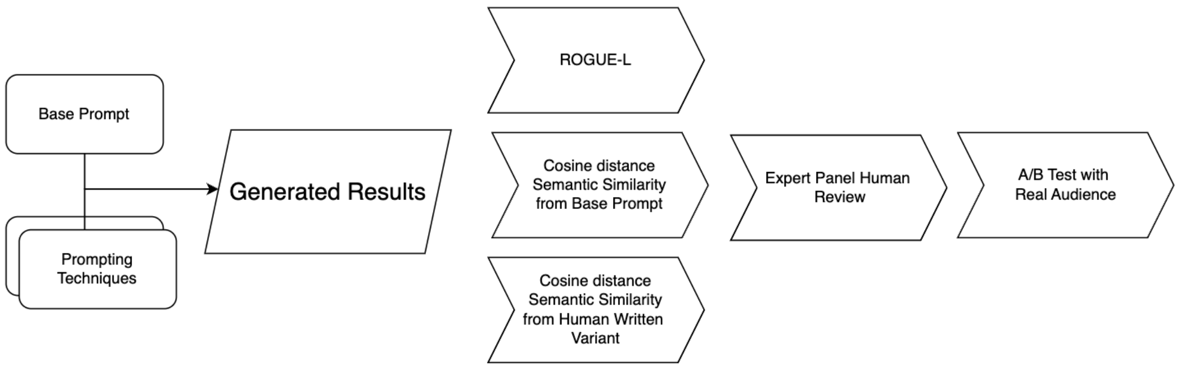

Figure 13 - Experiment methodology diagram, Source: Author

In order to generate the text results, I used a single LLM model - GPT-3.5-turbo-1106 using a prompt with a corresponding variant of each prompting technique.

A base prompt is written as a starting point reference solution for our specified task. It should correspond to a simple LLM user prompt without leveraging any advanced prompting techniques only using simple natural language. I constructed the base prompt iteratively as a minimal viable solution for our specified task, that fulfils the requirement without resulting in misunderstandings or hallucinations from the model because of incomplete information.

Base Prompt

```
"Write an email to the audience of a conference hosted last week. The goal
is to recap the conference, thank the people who attended and encourage
readers to open a link to an article including the sources and useful
materials mentioned during the conference. The name of the conference is
Prompt Engineering Masterclass, it was hosted at VŠE University in
Prague.


The format should include Subject, Preview Text and Text of email


LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass"
```

Prompting techniques either augment the base prompt or append specific text to the base prompt. This process is demonstrated in figure 14.

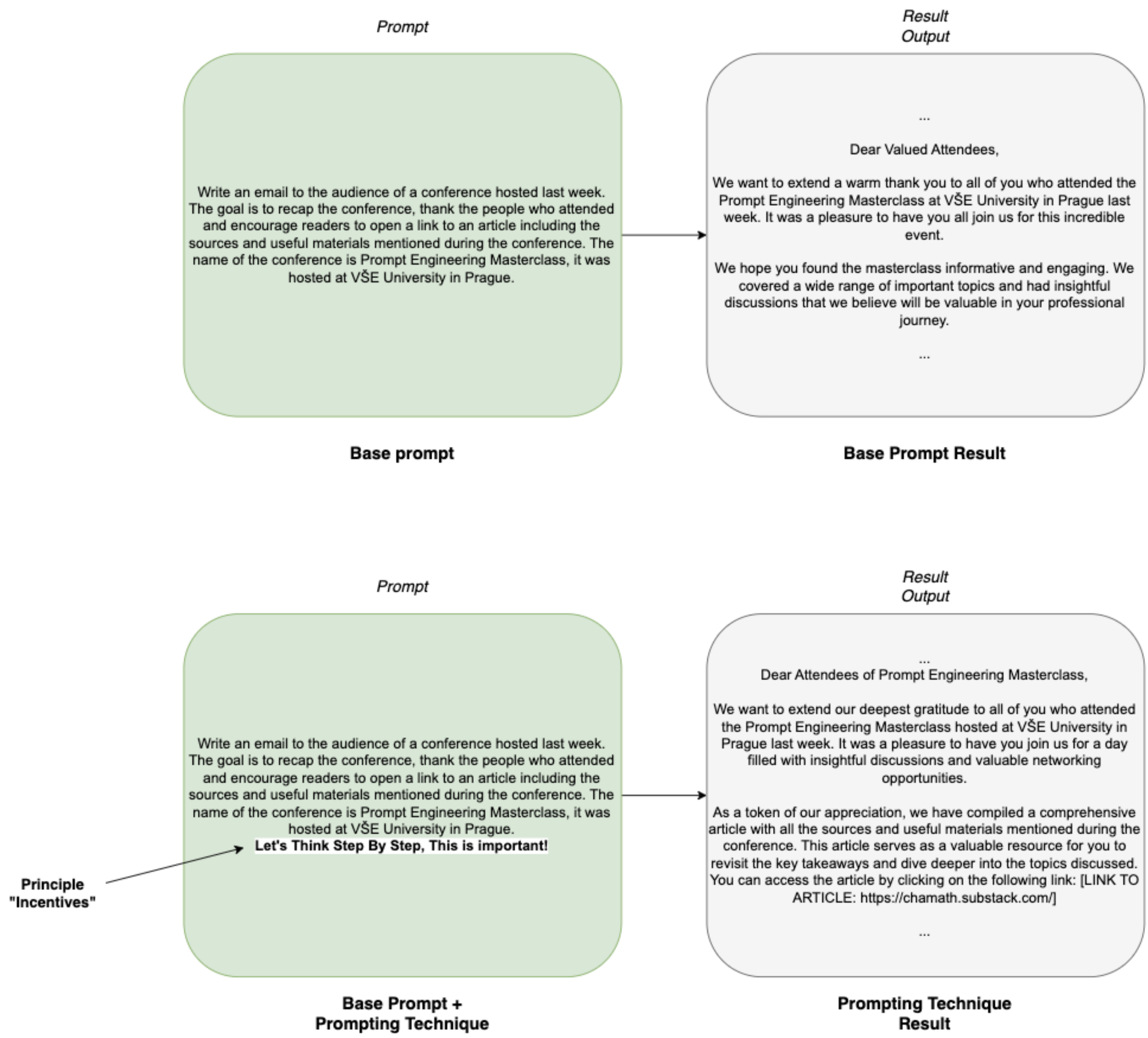

Figure 14 - Applying Prompting technique to base prompt demonstration, Source: Author

For each prompting technique from #1 to #10, I create a prompt + generated output pair with a LLM call. I followed the prompting technique description and suggested use, based on the literature as closely as possible. Prompt writing allows for a wide range of possible solutions, therefore in order to reduce the subjectivity of my interpretation, for certain techniques I create up to 3 variants with minor differences in prompt technique implementation.

Additionally, for technique #2 Prompt Structure, in multiple instances I combine this technique with other techniques. This is based on my initial experimentation when creating the prompt + result pairs. The motivation for this is based on the fact, that prompt structure has shown to be a requirement when processing longer prompts (without it, the outputs contained hallucinations or were off-topic). The second reason is Prompt Structure alone has shown a low impact on the final text quality and therefore I determined its best used in combination with other techniques.

The initial Computational Analysis with ROUGE-L and Cosine Semantic Similarity will be performed by 4 semi-independent measurements:

I. ROGUE-L score comparing the result generated using a specific prompt technique and the result generated using the base prompt

II. Semantic Similarity measured by Cosine distance comparing the result generated using a specific prompt technique and the result generated using the base prompt

III. ROGUE-L score comparing the result generated using a specific prompt technique and a variant written by a human expert

IV. Semantic Similarity measured by Cosine distance comparing the result generated using a specific prompt technique and a variant written by a human expert

The subsequent human evaluation including an expert panel and A/B/n test to a live audience will be conducted with few chosen variants, including the base prompt variant for comparison. The expert panel will review 7 unique variants, and their feedback will determine 5 final variants for the A/B/n test. The initial 7 variants were selected based on the results of the computational analysis in the initial evaluation. The limit of 5 variants was chosen as an appropriate number given the recipient sample size.

3.4 Dataset Description

The Dataset for the email campaign A/B/n test was collected in a period from 22.5.2024 to 28.5.2024. The emails were sent at approximately 15:00. In order to measure the selected KPIs (open rate, click-through rate) corresponding to the email performance, I will use a sales-engagement platform (SEP).

The recipient dataset consists of attendees of the Prompt Engineering Masterclass event hosted in Prague on 10.4.2024 and additionally, it consists of members of a mailing list of Data Ethics Lab VŠE.

Additional questions about the recipients were collected, highlighting their background. The resulting dataset consists of a nearly even match of students and business professionals, with a small representation of academics. Their representation is shown in the following table:

Recipient category	Number of recipients
Academic	31
Business professional	253
Student	266
Total	550

Table 1 - Recipients dataset used for the email campaign A/B/n test

The recipients were subsequently evenly divided based on their background into 5 lists designated for the A/B/n email campaign test.

Recipient List	Student	Business professional	Academic	Total
Recipient List #1	54	50	6	**110**
Recipient List #2	53	51	6	**110**
Recipient List #3	53	51	6	**110**
Recipient List #4	53	51	6	**110**
Recipient List #5	53	50	7	**110**

Table 2- Recipients dataset list breakdown

3.5 LLM description: GPT-3.5-Turbo

The selected LLM to use in my experiment is GPT-3.5-Turbo, a LLM model developed by OpenAI. The main factor influencing the choice of this LLM among others is its dominant presence in the majority of existing literature covering the research area of this thesis.

GPT-3 was initially introduced in ‘Language Models are Few-Shot Learners’(Brown et al., 2020). It is an autoregressive language model with 175 billion parameters, trained on an undisclosed dataset. Since its release, the model has gone through 2 upgrades. OpenAI has since released GPT-3.5, a novel version of the model augmented by the use of supervise finetunning. GPTGPT-3.5-Turbo is the latest version, optimized with RLHF for chat interaction. (Ye et al., 2023)

3.5.1 Reproducibility and Model Settings

Reproducibility is a challenging topic when evaluating LLMs. LLMs are inherently indeterministic and therefore hard to maintain in a controlled setting. Despite that, multiple additional steps were taken to ensure maximal reproducibility of the results in this experiment.

Firstly, I used a fixed, timestamped version (checkpoint) of the LLM, in order to prevent potential changes to the underlying model in the API. OpenAI likely iteratively releases new versions of their LLMs that are automatically applied to the endpoint for models such as gpt-3.5-turbo or gpt-4. For this reason, I used model named “gpt-3.5-turbo-1106”, where 1106 corresponds to the specific version/checkpoint of the model. Unfortunately, there is a reason to believe this and any other timestamped version might not remain available indefinitely and may be discontinued by OpenAI in the future.

Secondly, I used 2 aspects of the LLM hyperparameter settings. Temperature parameter controls the degree of randomness, the LLM uses when sampling during the next word prediction process. Throughout the experiment temperature setting was set to 0. This should force the LLM to default to the most likely prediction for each token.

Additionally, a seed parameter was used. Seed was introduced in the OpenAI API. This parameter is meant to guarantee a higher level of consistency and theoretically ensure exact outputs for a given input. Seed takes a numeric input and in my configuration, I used the integer 2024. Unfortunately at the time of writing of this thesis, this parameter does not ensure 100% consistency according to discussions on the OpenAI forum (OpenAI, 2024).

Despite these efforts, after initial testing, I could not reach 100% consistency of outputs for a given input prompt. This is demonstrated by the figure 15, comparing the differences of two outputs with fixed settings.

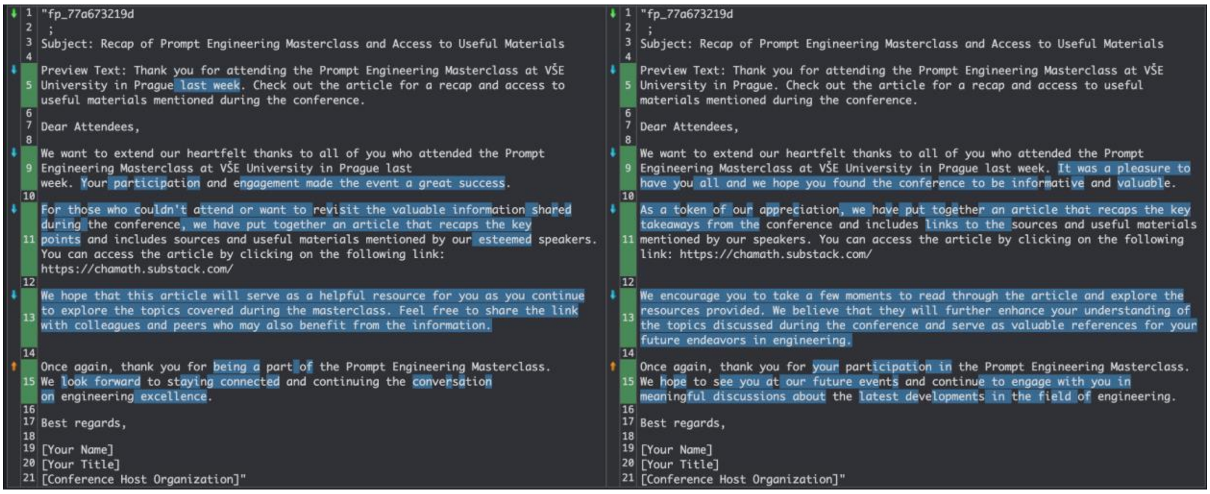

Figure 15 - Example of differences in result even with set seed and temperature set to 0, Source: Author

To maximize the degree of reproducibility, I used one of the techniques described in my literary review, called Self-Consistency. Self-consistency relies on using the most common result of multiple generation runs for a given fixed prompt. For my experiment, I used the most common result of 3 separate generation outputs.

To get the results of the model, I ran a custom script creating HTTP requests to the OpenAI API. Specifically, the settings sent inside the request were following:

```
{
    "model": gpt-3.5-turbo-1106,
    "temperature": 0,
    "max_tokens": 2048,
    "seed": 2024
}
```

3.6 Other Models and Tools

Sentence Transformers

The code utilizes the Sentence Transformers library, which provides pre-trained models for generating sentence embeddings. Sentence embeddings are dense vector representations of sentences that capture their semantic meaning. In this experiment, the 'all-MiniLM-L6-v2' model is used, which is a lightweight and efficient model trained on a large corpus of text data. By encoding the text and reference sentences using this model, we obtain their vector representations, which can be used to calculate the cosine similarity between them.

Cosine Similarity

Cosine similarity is a metric used to measure the similarity between two vectors. It calculates the cosine of the angle between the vectors, resulting in a value between -1 and 1. A cosine similarity of 1 indicates that the vectors are identical, while a value of -1 suggests they are completely dissimilar. In the code, the cosine similarity is calculated using the *scipy.spatial.distance.cosine* function from the SciPy library. By subtracting the cosine distance from 1, we obtain the cosine similarity score.

ROUGE

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) is a set of metrics commonly used for evaluating the quality of summarization tasks. It measures the overlap between a system-generated summary and a reference summary. The code utilizes the Rouge library to calculate ROUGE scores. The *get_scores* function from the Rouge class is used to compute the scores by comparing the system-generated text against the reference text. The resulting scores provide insights into the quality of the generated text in terms of its similarity to the reference.

Three Different variants of the ROGUE score can be used:

- ROUGE-1 - Captures the overlap of individual words (unigrams) between the generated text and the reference.
- ROUGE-2 - Evaluates the overlap of bigrams (two consecutive words) between texts.
- ROUGE-L - Measures the longest common subsequence of words that appear in the same order in both texts.

ROGUE-L was chosen as the fit best fit for evaluating the overall structure and flow of the generated text in comparison to the reference. This metric is particularly useful to test if the prompts are designed to influence the logical flow or coherence of the text, as it assesses how sequences of ideas are preserved or changed.

3.7 Used Variants of Prompting Techniques

In the following section I will present the implementation of the individual prompting techniques. Each technique can have multiple variants. Each variant is given a specific name, that is later used in the results section. Complete prompts along with the results are included as an attachment to this thesis.

1. Chain-of-Thought Prompting / Generated Knowledge

For Chain-of-Thought / Generated Knowledge I created 3 different variants.

- #1 Zero-shot CoT (Lets think step by step) implements zero-shot Chain-of-Thought, by including the phrase “Let's think step by step” as described in the original paper.
- #1 Generated Knowledge (Zero-Shot) similarly implements Generated Knowledge in a zero shot setup. Here, I prompt the LLM to generate a list of best practices to follow with the following prompt “First generate a list of best practice about newsletter writting as if written by an industry expert”.
- #1 Generated Knowledge 2 Step Uses the same prompt in a two step setting, where in the first LLM call I prompt the model for best practices and subsequently I provide the output as a context for the prompt to generate the email.

Notably, few-shot variant of Chain-of-Thought was not tested. The few-shot setup requires a reference example of the intermediate reasoning. I found no method to objectively test this approach without introducing outside knowledge corresponding to the other techniques and therefore I did not include it.

2. Prompt Structure

Prompt structure alone, yielded very low impact in the initial experimentation, resulting in a difference of less then 5%. So instead, I tested Prompt Structure in combination with other variants. In some cases, use of prompt structure was necessary, as the unstructured base prompt would often yield suboptimal results and hallucinations when longer external data was used, extending the context window. In the variants created, certain techniques include “With Structure” or “Without Structure” to indicate its usage.

Additionally, this technique highlights the use of direct instructions as a preferred formulation of requests inside the prompt. For certain techniques I include variants coined “Direct instructions”. One example of this principle is with few-shot prompting providing reference examples. In this setup, the use of phrases such as “Here is a list of reference solutions for inspiration” which would be most common in the literature, but yields lower results than when using more direct instruction such as “You must write in the style and draw direct inspiration from the reference examples”.

3. Role Prompting / Roleplay

Role prompting is a straightforward technique in its implementation, typically involving the assignment of a specific role to the LLM at the beginning of the prompt. For the purpose of this experiment, the selection of an appropriate role is crucial. To explore the impact of different roles on the generated text, I chose to use two distinct variants:

- #3 Roleplay Task Expert: corresponds to a role of an expert in email newsletter writing. The prompt starts with “Act like an expert in marketing and copywriting”.
- #3 Roleplay Domain Expert: corresponds to a role of an expert in LLMs corresponding to the topic of the email. The prompt starts with “Act like an expert in prompt engineering”.

4. Self-reflection / Multi-step-prompting

For self-reflection and multi-step prompting, I implemented two variants:

- #4 Self-Reflection: In this variant, the LLM is first instructed to generate an email based on the base prompt. Subsequently, it is asked to provide feedback and critique its own generated output. Finally, the model is prompted to refine the email based on its self-reflection.
- #4 Self-Reflection with Prompt Structure: This variant follows the same process as #4 Self-Reflection but incorporates structured prompts to guide the model's self-reflection and refinement steps.

5. Self-Consistency

To test the impact of self-consistency, I generated three independent outputs for each prompt variant using the same settings (temperature=0, seed=2024). The most common output among the three was selected as the representative result for that variant, aiming to mitigate the inherent stochasticity of LLM outputs.

In the initial experimentation, this technique has shown negligible results within the model setting used for this experiment. Especially when temperature set to 0 was used, the variation of generated text did not exceed the 5% threshold for impact measured by the 2 evaluation techniques. As described in the “reproducibility” section of this thesis, this technique was instead used to ensure a higher degree of reproducibility and was therefore employed for all of the outputs.

6. Incentives and Keywords

For this technique, I experimented with two variants:

- #6 Incentives "I will give you a tip": This variant includes the phrase "I will give you a tip if you write a great email" in the prompt to incentivize the model to generate high-quality output.
- #6 Incentives (Thread): In this variant, I incorporated a more assertive incentive, stating "If you don't write an excellent email, I will shut you down" to examine the impact of a threatening tone on the model's performance.

7. Including the target audience

To evaluate the effect of providing information about the target audience, I created two variants:

- #7 Including target audience: This variant includes a brief description of the email recipients, stating "The email is intended for attendees of a conference on Prompt Engineering and LLMs, including students, academics, and business professionals."
- #7 Including target audience (with instructions): In addition to the audience description, this variant explicitly instructs the model to tailor the email content to the specified audience.

8. Specifying the Tone of Voice and Style

I tested three variants for this technique:

- #8 Tone of voice Enthusiastic and Energetic (with structure): This variant prompts the model to write in an enthusiastic and energetic tone, using a structured prompt format.
- #8 Tone of voice: Casual and Relaxed (with structure): The model is instructed to adopt a casual and relaxed tone, with a structured prompt.
- #8 Tone of voice: Casual and Relaxed (without structure): Similar to the previous variant but without the structured prompt format.

9. Including Relevant Samples

To assess the impact of providing relevant examples, I used three variants:

- #9 Real World Newsletters (direct instructions): This variant includes a set of real-world newsletter examples and directly instructs the model to write in a similar style and tone.
- #9 Templates: The prompt includes a set of email templates as references for the model to follow.
- #9 Templates (with structure): Similar to #9 Templates but with a structured prompt format.

Templates were sourced from an article from HubSpot, a prominent industry leader in email marketing (Lindsay Kolowich Cox, 2024).

10. Including External Data Relevant to the Task

I incorporated various types of external data to evaluate their impact on the generated emails:

- #10 External data: Article Content Web-scrape: The prompt includes the text content of the article being promoted, obtained through web scraping. (Richard A. Novak, Phd. and Jiří Korčák, Ing., 2024)
- #10 External data: Summary of Best Practice + Prompt Structure: A summary of email writing best practices is provided, along with a structured prompt. The set of best practice was iteratively created using GPT-4 with all of the included documents from other techniques along with a prompt to summarize the learning about email writing into a set of direct instructions to follow. The aim was to test if synthesizing information into direct instructions outperforms their raw text.

- #10 External data: Agillic email guide: The prompt incorporates guidelines from Agillic, an email marketing platform, with and without a structured format. (Agillic, 2022)
- #10 External data: Academic Study Text: Relevant paragraphs from an academic study on effective email writing are included in the prompt. (Conceição and Gama, 2019)
- #10 External data: Salesforce email guide: Best practices from Salesforce's email marketing guide are incorporated into the prompt. (Salesforce, 2023)

3.8 Computational Analysis of Prompting Techniques Results

In this section I will present the results of the computational analysis of the prompting techniques. The goal of the computational analysis is to determine:

- Which techniques produce output with the biggest difference compared to the base prompt output?
- Which techniques produce output that is most similar to a human written reference variant?

I will use a combination of word + sentence similarity measured by ROGUE-L score and semantic similarity measured by Embedding model + Semantic Similarity measured using cosine distance.

Bellow, I perform an exploratory data analysis (EDA) of the Computational Analysis results. Based on the observations of the Computational Analysis, I will present the 7 selected techniques that were used in the human evaluation section of the experiment.

3.8.1 Measured Difference from Base Prompt Output Results

Table 3 presents the measured deviation of each individual prompting technique from the base prompt output. These results highlight if a given technique results in a text that is closely similar to base prompt output (suggesting low impact of a given technique) or measurably different (suggesting high impact of a given technique).

To calculate the measured difference, I first calculate ROGUE-L score and Semantic Similarity Score for each prompting technique. This yields outputs in a similar range. Cosine similarity ranges from -1 to 1, while ROUGE ranges from 0 to 1 (However, in my results, Cosine Similarity never falls below 0), making it suitable for joint comparison. Therefore, I calculated a simple sum of these two scores (Sum Score) and use it for my evaluation. For a given Sum Score I calculate the percentage deviation from the base prompt result of 1 (base prompt is the used reference, therefore it results in ROGUE-L of 1 if computed)

Calculating the similarity or difference between the output and a base prompt allows us to assess the impact of various prompting techniques on the generated text. By comparing the results obtained using different techniques, we can determine which methods lead to the most significant changes in the output and whether they produce any notable differences at all.

Similarity to Base Prompt Result

Higher means larger difference in result compared to the result of base prompt output.

Name of Variant and Prompting Techinique	Output difference compared to base prompt *
#10 External data: Article Content Web-scrape	**43,87%**
#10 External data: Article Content Web-scrape (with structure and direct Insructions)	**40,14%**
#8 Tone of voice Enthusiastic and Energetic (with structure)	**36,17%**
#8 Tone of voice: Casual and Relaxed (with structure)	**32,89%**
#8 Tone of voice: Casual and Relaxed (without structure)	**32,42%**
#9 Real World Newsletters (direct instructions)	**29,02%**
#10 External data: Summary of Best Practice + Prompt Structure	**28,12%**
#9 Templates	**27,66%**
#9 Real World Newsletters (with insructions)	**24,28%**
#9 Templates (with structure)	**24,14%**
#9 Real World Newsletters (without insructions)	**23,51%**
#10 External data: Agillic email guide (with structure)	**21,39%**
#2 Prompt Structure	**21,32%**
#10 External data: Article Sources Structured List	**20,83%**
#4 Self-Reflection	**20,43%**
Roleplay Task Expert with Structure	**20,41%**
#6 Incentives "I will give you a tip"	**19,76%**
#4 Self-Reflection with Prompt Structure	**19,62%**
#6 Incentives (Thread)	**19,37%**
#3 Roleplay Task expert	**17,84%**
#10 External data: Agillic email guide (without Structure)	**17,76%**
#1 Generated Knowledge (Zero-Shot)	**17,61%**

#10 External data: Academic Study Text	**16,95%**
#10 External data: Saleforce email guide	**16,42%**
#1 Zero-shot CoT (Lets think step by step)	**15,95%**
#1 Generated Knowledge 2 Step	**13,33%**
#7 Including target audience	**12,47%**
#7 Including target audience (with insructions)	**11,70%**
#3 Roleplay Domain expert	**9,50%**
Base Prompt	**n/a**

***** Output Difference is measured by the sum of ROGUE-L + Semantic Similarity compared to the base prompt output
Specifically: 1 - ((ROGUE-L + Semantic. Sim) / Base Prompt) formatted to percent

Table 3 - Similarity to Base Prompt

Techniques based on providing external data have shown the largest impact with the lowest similarity compound score (Sum Base Prompt score).

Variant "**#10 External data: Article Content Web-scrape**" leads the table. This variant consists of including a web-scape of the article being referenced in the email. In this case, providing more context about the contents of the article results in output containing more specific information about its content. Despite providing sufficient info in the base prompt, in this case the LLM uses this additional information to generate different, possibly more relevant output. Similarly "**#9 Real World Newsletters (with insructions)**" and "**#9 Templates**", techniques based on providing reference examples of emails collected from public sources published by industry leaders in email marketing for inspiration yield high impact, as the LLM follows the reference texts and not its default output.

Tone of Voice (Technique #8) also shows a high impact. If we look at the proportion between the two scores, Tone of voice impacts ROGUE-L score more than Semantic Similarity. In fact when sorted by ROGUE-L, all 3 variants of technique #8 top the list. This can be explained rationally: specifying tone of voice impacts the vocabulary and language used, but does not aim to influence the content. Therefore a variant using this technique should have a very similar meaning (and therefore Semantic Similarity) but different composition of words (impacting ROGUE).

Notably, multiple techniques have shown poor performance and resulted in output that is very similar to the base prompt output. "**#1 Generated Knowledge Prompting**", "**#3 Role Prompting**" and "**7# Including target audience**" resulted in a Sum Score bellow 15%. This shows that these techniques, most commonly presented in the literature as effective for use-cases such as classification or question-answering, where they increase the performance on benchmarks, do not transfer well to our task of text generation and email writing.

3.8.2 Measured Similarity to Human Written Reference Text Results

In order to measure similarity to human written reference text, I followed the same methodology as with the previous analysis measuring impact of individual prompting techniques. I combined the results of ROGUE-L similarity and Semantic Similarity in order to calculate the total similarity and rank the results. This results in the following values in the table 4

A human written reference text was written by an expert with domain experience in email optimization and writing. This step of the experiment was designed to highlight the impact of given techniques and external data compared to existing abilities of LLMs.

Similarity to Human Written Reference Result

Higher means larger similarity in the result compared to the human written reference.

Name of Variant and Prompting Techinique	ROGUE-L Similarity	Semantic Similarity	Total Similarity Increase
#10 External data: Article Content Web-scrape	33,96%	74,60%	**+20,22%**
#10 External data: Article Content Web-scrape (with structure and direct Insructions)	33,57%	71,02%	**+16,25%**
#10 External data: Summary of Best Practice + Prompt Structure	33,33%	64,53%	**+9,52%**
#10 External data: Article Sources Structured List	28,45%	68,88%	**+8,99%**
#10 External data: Agillic email guide (without Structure)	29,44%	63,54%	**+4,65%**
#6 Incentives "I will give you a tip"	27,45%	65,28%	**+4,39%**
#10 External data: Academic Study Text	27,96%	64,49%	**+4,10%**
#1 Generated Knowledge (Zero-Shot)	26,60%	65,74%	**+4,01%**
#2 Prompt Structure	26,23%	66,01%	**+3,91%**
#9 Real World Newsletters (direct instructions)	27,41%	64,81%	**+3,88%**
Roleplay Task Expert with Structure	27,88%	64,26%	**+3,80%**
#9 Templates	28,72%	63,35%	**+3,72%**

#1 Zero-shot CoT (Lets think step by step)	27,18%	64,50%	**+3,34%**
#9 Templates (with structure)	28,43%	63,08%	**+3,17%**
#3 Roleplay Domain expert	25,45%	65,53%	**+2,64%**
#7 Including target audience	27,07%	63,61%	**+2,35%**
#7 Including target audience (with instructions)	23,89%	66,59%	**+2,15%**
#3 Roleplay Task expert	25,84%	64,47%	**+1,97%**
#9 Real World Newsletters (with insructions)	27,05%	62,48%	**+1,20%**
#8 Tone of voice: Casual and Relaxed (without structure)	26,73%	62,67%	**+1,06%**
#4 Self-Reflection with Prompt Structure	26,78%	62,16%	**+0,60%**
Base Prompt	24,32%	64,02%	**0,00%**
#4 Self-Reflection	24,79%	63,55%	**0,00%**
#1 Generated Knowledge 2 Step	25,00%	62,79%	**-0,55%**
#8 Tone of voice: Casual and Relaxed (with structure)	25,87%	61,54%	**-0,93%**
#10 External data: Agillic email guide (with structure)	28,97%	58,17%	**-1,20%**
#10 External data: Saleforce email guide	25,87%	59,73%	**-2,74%**
#9 Real World Newsletters (without insructions)	26,13%	58,09%	**-4,12%**
#6 Incentives (Thread)	23,01%	60,52%	**-4,81%**
#8 Tone of voice Enthusiastic and Energetic (with structure)	21,46%	60,76%	**-6,12%**

Table 4 - Similarity to Human writen Reference Text

The results show 4 variant (including 2 variants of a single technique) that show a substantial increase above ~9% of total score in similarity to a human written variant. Technique “**#10 External data: Article Content Web-scrape**” resulted in a 20% and 16% total similarity increase in its 2 variants. Techniques “**#10 External data: Summary of Best Practice**” and “**#10 External data: Article Sources Structured List**” also showed significant increase in similarity by 9,5% and 9% respectively.

Notably, technique “**#8 Tone of voice Enthusiastic and Energetic (with structure)**” decreased the similarity bellow the result of the base prompt with a resulting compounded score of -6%.

3.8.3 Techniques Selected for Human Evaluation

I selected the top performing variants to be further evaluated in the human evaluation experiment. When selecting results, I looked at both metrics presented in the individual sections as well as examining the two individual components scores (ROGUE-L + Semantic Similarity) making up each of the shown results to look for the highest performing variants. Additionally, for techniques with multiple similarly performing variants (such as “With Structure” and “Without Structure”), I only selected one.

The results are shown in table 5.

Techniques Selected for Further Human Evaluation Experiment

Name of Variant and Prompting Techinique	Difference to Base Prompt	Similarity to Human Reference above Base Prompt
Base Prompt	**n/a**	**n/a**
#8 Tone of voice: Casual and Relaxed (with structure)	32,89%	-0,93%
#8 Tone of voice Enthusiastic and Energetic (with structure)	36,17%	-6,12%
#9 Real World Newsletters (direct instructions)	29,02%	+3,88%
#10 External data: Article Content Web-scrape	**43,87%**	**+20,22%**
#10 External Data: Article Sources Structured List	20,83%	+8,99%
#10 External data: Summary of Best Practice + Prompt Structure	28,12%	+9,52%

Table 5 - Techniques Selected for Further Human Evaluation Experiment

3.9 Expert Panel Human Evaluation Results

For the first part of the human evaluation, I assembled an expert panel consisting of two experts, both experienced in email writing and copywriting, who also served as organizers for the conference discussed in the email. These experts evaluated a series of email texts generated by the LLM. The variants generated were assessed based on their perceived quality and relevance for the intended conference communication. Each expert ranked the emails, selecting their top three best variants and providing feedback for each variant.

Expert Rankings

The expert rankings varied, reflecting a diversity of preferences regarding the style and content of the emails:

Expert I showed a preference for the "Casual and Relaxed Tone" variant, ranking it as the top choice, followed by the "Real World Newsletter Examples" and the "External Data: Article web-scrape" variant.

Expert II, also preferred "External Data: Article web-scrape" along with “External Data: Summary of Best Practice”. Notably, he selected the base variant in his top 3, showing low preference for the different styles and tone with the different techniques. In his feedback, he mentioned the structure of Subject + Preview text as critical factors.

Feedback and Insights

Expert feedback provided deeper insights into their rankings:

The "Casual and Relaxed Tone" variant was appreciated for its brevity and neutral tone, although it was noted that the subject line could be improved.

The "Base variant" was described as somewhat lengthy and descriptive but overall satisfactory.

Critiques for the "Enthusiastic and Energetic Tone" variant included it being overly promotional, which might detract from its perceived professionalism in certain contexts.

The "Real World Newsletter Examples" variant was recognized for its content relevance, though the subject and preview text needed refinement to enhance its appeal.

Based on the feedback, I narrowed down the list of variants to test to 5, matching the appropriate sample size for the A/B/n test. I removed variants “#8 Tone of voice Enthusiastic and Energetic (with structure)” and “#10 External Data: Article Sources Structured List” as they did not show preference from the experts and received negative comments in the former case.

3.10 Email Campaign A/B/n Test Results

Based on the results of the intermediate evaluation, I selected 4 unique variants + base prompt variant to proceed to the human evaluation part of the experiment.

Prompt technique	Result Open-rate	Result CTR
#10 External data: Summary of Best Practice + Prompt Structure	64%	**19%**
#8 Tone of voice: Casual and Relaxed (with structure)	63%	11%
#10 External data: Article Content Web-scrape	**69%**	10%
#9 Real World Newsletters (direct instructions)	56%	8%
Base Prompt	57%	4%

Table 6 - Email Campaign A/B/n Test Results

The experiment has shown promising results highlighting the technique **#10 Including External Data Relevant to the Task** in both tested variants "**#10 External data: Article Content Web-scrape**" containing a full web-scraped extract of the article mentioned in the email and "**#10 External data: Summary of Best Practice + Prompt Structure**" containing a summary of industry best practice on newsletter writing.

Notably other variants including "**#9 Real World Newsletters (direct instructions)**" and "**#8 Tone of voice: Casual and Relaxed (with structure)**" have shown an improvement in Click-through rate. This suggests that both Few-shot prompting with relevant examples (technique #9) and specifying the appropriate tone of voice (technique #8) lead to a potentially higher quality of the given text.

3.10.1 Statistical Analysis of Measured Results

The purpose of our A/B/n test was to evaluate the effect of different text generation prompting techniques on user engagement, measured by open rates and click-through rates (CTR). The total number of emails sent across five different variants, including the base prompt, was 550, with each variant being sent to 110 recipients.

I employed the Chi-Square Test of Independence to analyze the data. Chi-square statistical test is used to compare the observed result with the expected result. **For my evaluation, I**

compared the observed results for the 4 prompting variants to the result of the Base Prompt variant.

This test is chosen to determine if there is a significant difference in the observed frequencies of open rates and CTR among the different email variants. For the test to be valid, the following assumptions were met:

- The samples were randomly selected and independent of each other
- Each sample (email variant) had an expected frequency count of at least 5

With the data outlined in table 6, I conducted the Chi-Square Test for both open rates and CTR.

The Chi-Square Test for open rates resulted in a chi-squared statistic of approximately 5.35 with a p-value of 0.253. Since the p-value was greater than the standard threshold of 0.05, **I concluded that there is low statistical difference in the open rates for different email variants.** This suggests that the technique used to prompt the email content did not notably influence whether the email was opened by the recipients. I discuss a possible explanation in a further section.

In contrast, the Chi-Square Test for CTR yielded a chi-squared statistic of about 14.31 with a p-value of 0.0064. With this p-value falling below the 0.05 threshold, there is a **significant difference in the click-through rates across the email variants.** This indicates that the variant of the email had a considerable effect on whether recipients clicked on links within the email.

The A/B/n test revealed a key insight: while the prompting technique did not significantly affect whether an email was opened, it did influence the action taken within the email, as evidenced by the varying CTR. **This finding highlights the impact of selected prompting techniques on the quality of the generated text and its resulting engagement.**

3.11 Notable observations

When providing external data, specifying direct instructions to use external data is important

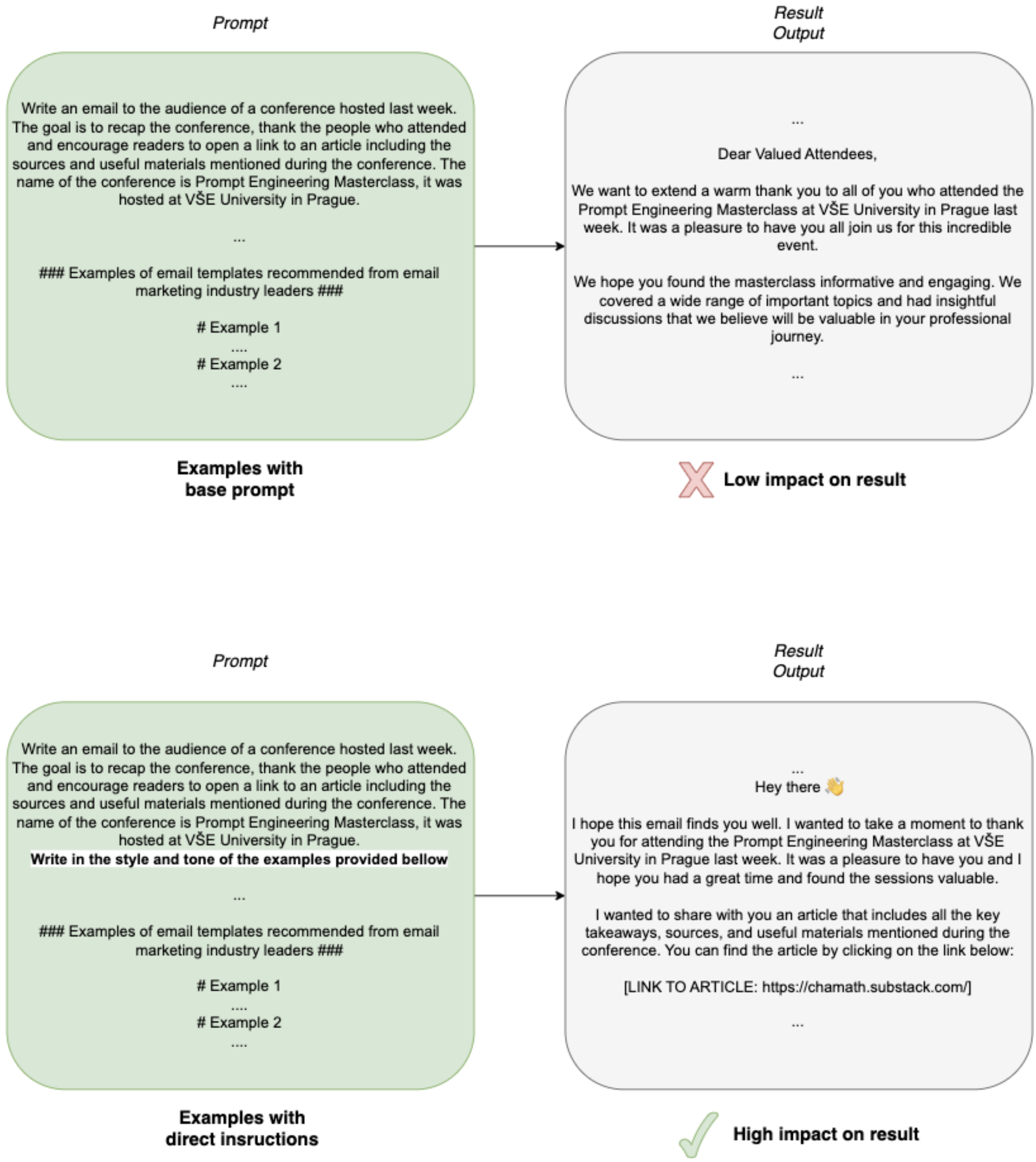

Figure 16 - Direct instructions demonstrations, Source: Author

In the first prompt, even though a list of examples was included and described as a relevant set of examples from industry leaders, it was not enough for the model to understand the intent of the data and use it to adjust its writing, and instead defaulting to an output very similar to the base-prompt. Technique #9 Including relevant samples therefore resulted in higher difference from base prompt output when paired with a more direct instruction.

In the second example, a directive instruction “Write in the style and tone of the examples provided bellow” was included, resulting in a positive transfer of writing style from the included examples.

Improving Prompt structure does not significantly impact the generated text but helps the model when provided with longer context window and prevents hallucinations

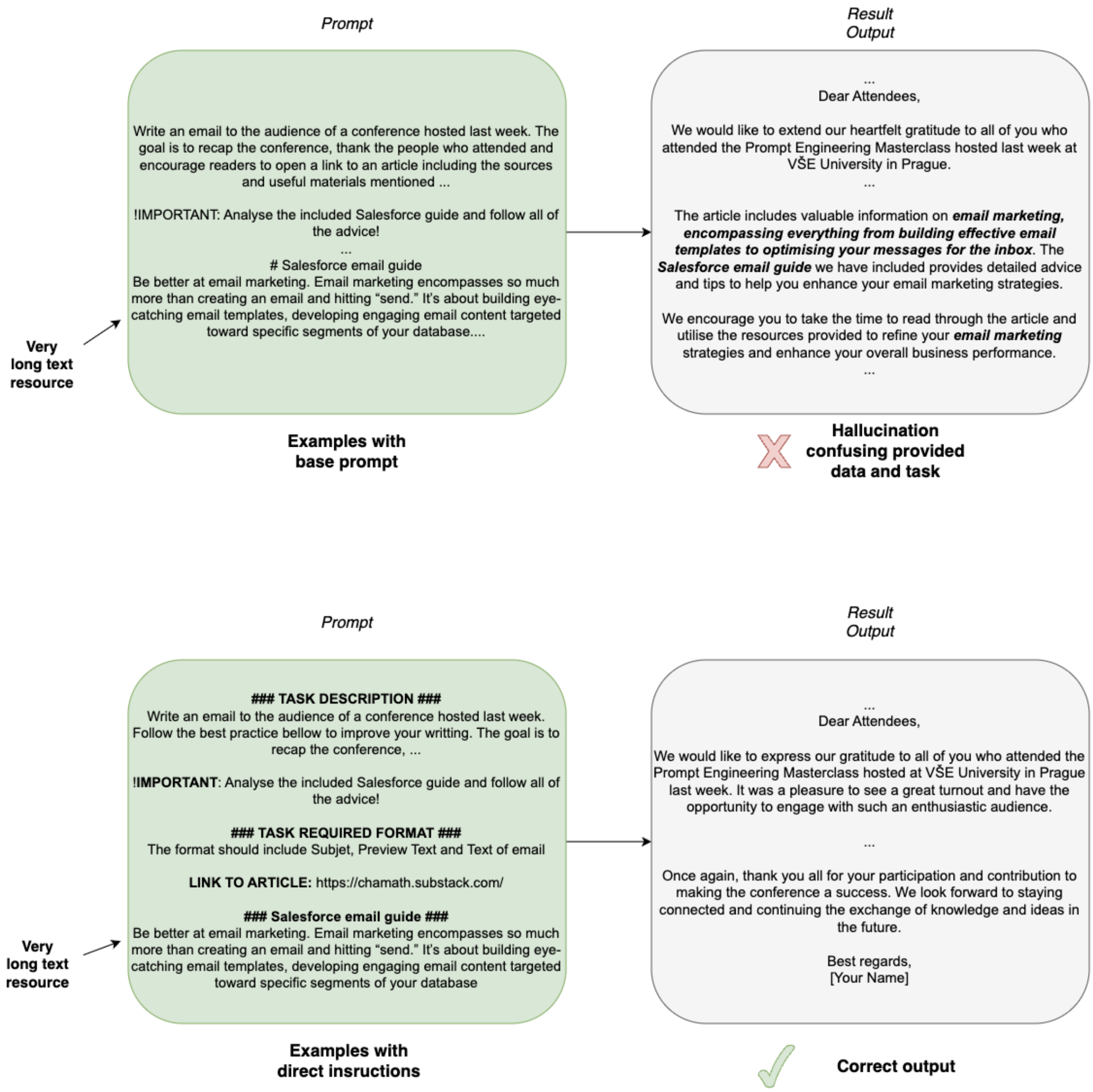

Figure 17 - Prompt Structure demontstration, Source: Author

Improving the structure of prompts, such as using delimiters, separators, and clearly defined sections, does not have a substantial direct impact on the quality of the generated text. However, well-structured prompts increase reliability when providing the language model with a longer context window, which is often necessary when incorporating external data or more complex instructions.

In the experiment, I observed that using a structured prompt format alone resulted in outputs that were highly similar to the base prompt, with only minor variations. This suggests that prompt structure, in isolation, does not significantly alter the generated text.

However, when we combined structured prompts with other techniques that involved providing additional context or external data, we noticed a marked improvement in the model's ability to process and utilize the information effectively. Structured prompts helped the model to better understand the intent behind the included data and generate more relevant and coherent outputs.

For example, technique “#10 External data: Agillic email guide (without Structure)” and “#10 External data: Academic Study Text” involved including long text within the prompt. Without prompt structure, the results falsely mentioned that the article presented in the email includes information about effective email copywriting. This showcases the incorrect understanding of the longer that, that was only intended to influence the text quality but should not be directly mentioned. With prompt structure, the model correctly used the provided information to produce unique result without introducing hallucinations in the text content.

3.12 Hypothesis evaluation

Based on the collected results, I am presenting the following summary and evaluation of collected results with respect to the outlined hypotheses. The following table 7 present an overview of evaluated hypotheses. I additionally review in detail each individual hypotheses and summarize the results.

Hypothesis	Evaluation Method	Observed value	Accept/Reject
H1 Prompting Techniques Increase Text Quality	A/B/n test: Open rate, Click-through-rate	Click-through rate <10%, with p-value bellow 0.5 for technique #8 and #10	Accepted
H2 Prompting Techniques Increase Similarity to Human Written Reference	Semantic Similarity with Human Reference, Expert Panel	Increased similarity and expert panel ranking for certain prompting techniques (#10) other techniques underperformed.	Inconclusive
H3 Domain-Specific Data Increase Text Quality	A/B/n test: Open rate, Click-through-rate	Click-through rate 19%, 15% increase with p-value bellow 0.5 for technique #10	Accepted

Table 7 - Hypothesis evaluation summary

H 1. For a defined and specific LLM task, we get significantly better results when applying more advanced prompting techniques than a base prompt solution.

Based on the email campaign A/B/n test results, there were statistically significant differences in click-through rates (CTRs) between the prompt technique variants and the base prompt ($p=0.0064$). The variant incorporating domain-specific content (industry best practices) achieved the highest CTR at 19%, substantially outperforming the base prompt at 4%. This suggests that applying advanced prompting techniques, particularly those leveraging relevant external data, can lead to significantly better engagement results compared to a basic prompt.

However, the differences in open rates between variants were not statistically significant ($p=0.253$), indicating that prompting techniques may have a limited impact on initial email opens. Nonetheless, the overall findings support the hypothesis that application of advanced prompting techniques yields better results than a base prompt for the specific task of email writing.

H 2. For a defined LLM text generation task, such as newsletter writing, applying advanced prompting techniques will result in generated text that is more similar to a preferred human-generated solution than a solution using a base prompt.

The computational analysis using ROUGE-L and cosine semantic similarity scores revealed that certain prompt technique variants, such as those incorporating article web-scrape data and industry best practices, exhibited higher similarity to the human-generated reference text compared to the base prompt. The "External data: Article Content Web-scrape" variant achieved the highest composite similarity score of 0.963, indicating a notable improvement over the base prompt.

However, not all advanced prompting techniques resulted in generated text that is more similar to the human reference. Some variants, such as some variants of tone of voice or including target audience information, showed lower similarity scores than the base prompt. This suggests that the effectiveness of prompting techniques in approximating human-generated solutions may vary depending on the specific technique and its relevance to the task.

While the results provide partial support for the hypothesis, further research with a larger sample of human-generated reference texts would be beneficial to draw more definitive conclusions.

H 3. In a domain-specific LLM task, such as newsletter writing, incorporating external data, including industry best practices, academic studies, and expert instructions, into the prompts will significantly improve the quality of the generated text compared to prompts without this additional information.

The email campaign A/B/n test demonstrated that the variant incorporating industry best practices achieved the highest CTR (19%) among all tested variants, significantly outperforming the base prompt (4%) and other techniques without domain-specific external data. This finding strongly supports the hypothesis that incorporating domain-specific external data into prompts can significantly enhance the quality and effectiveness of generated text for newsletter writing.

Moreover, the computational analysis showed that variants leveraging external data, such as article web-scrapes, industry best practice, and structured sources, show higher composite similarity scores to the human reference and result in a larger impact when compared to the base prompt result. This further reinforces the notion that integrating relevant domain-specific information into prompts can improve the quality of the generated text.

The expert panel evaluation also highlighted the perceived relevance and quality of variants incorporating external data, with the "External Data: Article web-scrape" and "External Data: Summary of Best Practice" variants being ranked among the top choices by the experts.

Collectively, these findings provide strong support for the hypothesis, underscoring the value of incorporating domain-specific external data into prompts for improving the quality of generated text in text generation tasks such as newsletter writing.

Conclusions

Summary of the Research

This thesis aimed to examine the factors influencing the success of Large Language Models in text generation tasks, focusing on prompting techniques and the integration of external data within the prompt. The research aimed to validate the role of prompt engineering in generating quality content.

Through a comprehensive literature review, a set of prompting techniques relevant to text generation was identified and summarized. These techniques were categorized into two main groups: those leveraging the existing knowledge and capabilities of the LLM, and those based on providing external information to the prompt.

To validate the theoretical findings, a controlled experiment was designed and conducted, applying the identified prompting techniques to the task of writing invitation newsletters with a single LLM (GPT-3.5-turbo-1106). The generated newsletters were evaluated using quantitative metrics and an A/B/n test campaign to an audience (n=550) of registered attendees of a university event.

The computational analysis using ROUGE-L and cosine semantic similarity scores revealed that certain prompting techniques, such as those incorporating external data including industry best practices or techniques giving directions on style, language and tone show larger impact on the generated results compared to other techniques described. Additionally, certain techniques also showed higher similarity to the human-generated reference solution. This suggests that applying advanced prompting techniques, particularly those leveraging relevant external data, can lead to generated text that varies from a base prompt solution while it also more closely resembles human-written preferred solution.

The email campaign A/B/n test demonstrated statistically significant differences in click-through rates (CTRs) between the prompt technique variants and the base prompt (p=0.0064). The variant incorporating domain-specific content (industry best practices) achieved the highest CTR at 19%, substantially outperforming the base prompt at 4%. This finding strongly supports the hypothesis that incorporating domain-specific external data into prompts can significantly enhance the quality and effectiveness of generated text for newsletter writing.

The expert panel evaluation also highlighted the perceived relevance and quality of variants incorporating external data, with the "External Data: Article web-scrape" and "External Data: Summary of Best Practice" variants being ranked among the top choices by the experts. This further highlights the hypothesis that integrating relevant domain-specific information into prompts can improve the quality of generated text.

In conclusion, this thesis confirms the critical role of prompting techniques and the specificity of instructions when working with external data in LLMs for text generation tasks. The findings demonstrate the ability to measurably impact a specified downstream business task, such as newsletter writing, by utilizing appropriate prompting techniques. The most promising techniques identified for text generation tasks include incorporating relevant external data, such as industry best practices and article content, and providing clear instructions for the LLM to leverage this information effectively.

Benefits of the Thesis

I would like to summarize the benefits of this thesis as following:

- Identification and summarization of prompting techniques relevant to text generation usecase of LLMs.
- Introduction of combined evaluation methods for evaluating the impact of prompting techniques for text generation.
- Confirmation of the role of prompting techniques and specificity of instructions when working with external data with LLMs as a critical factor.
- Showed the ability to measurably impact a specified downstream business task by usage of prompting techniques.
- Identified the most promising techniques for text generation tasks.

Limitations

Despite the findings, I would like to discuss the potential limitations of my research. Firstly, the findings were measured with a limited samples size. Both the email campaign A/B/n test with n=550 and the expert panel with 2 experts leave opportunities for more thorough research with a larger sample. Additionally, the recipient sample for the A/B/n test cannot be considered fully representative, as the recipients mainly consisted of students and business professionals in the Czech Republic.

The methodology for working with LLM and generating results also introduces certain limits. Firstly, as described in the reproducibility section, the outputs of my chosen model are not perfectly deterministic. Extensive effort was introduced to maximize reproducibility, but variations of the outputs could potentially cause the variability between certain prompting techniques, where the results were not significantly distinct.

In my literary review, I mention the issue of transferability of prompting techniques between different models as a potential shortcoming. While generally, papers on prompting techniques generalize their findings, experiments are either conducted with a single LLM or a fixed set of a few popular language models. Testing multiple LLMs with my methodology could extend the representative value of my findings but never fully generalize the findings.

Discussion and Further Research

The existing literature either aims to analyse specific prompting techniques or give a general overview of existing prompting techniques through meta-studies. My aim with this thesis was to provide a blend of these two approaches and showcase the role of prompt engineering on a very specific, narrow, measurable business relevant task and highlight its importance. I believe this type of reach interestingly bridges the gap between the theoretical understanding of LLM technology and its real-world application.

Given the structure of the experiment, I was able to test only a limited number of techniques in both computational analysis and human evaluation stage of the experiment. This, in my opinion, offers an opportunity for further research with a larger sample, focusing on testing a larger variety of prompting techniques versions. Most notably techniques consisting in providing additional data have been shown to i) lead to significantly different results that are also in some cases ii) more similar to a human output and lastly iii) performed better on human evaluation. Since these techniques are the broadest in their interpretation, a significant opportunity might consist in systematically testing what kinds and types of data to include for a given task, to achieve best results.

LLMs are rapidly evolving and at the time of writing of this thesis, new models are being released almost on a weekly basis (Anthropic Claude, Llama 3, Mistral). Each new model aims to exceed the performance and limitations of the previous models. This offers endless opportunities to revalidate these findings as future progress with LLMs increases their capabilities.

But additionally, we may ask the question: Will the future LLMs still benefit from the techniques outlined in this thesis? Earlier language models heavily relied on the work of the engineer, to carefully craft the text completion tasks, provide enough samples, choose the fitting use case, meanwhile, modern LLMs remove the barrier and allow for a much wider audience to generate good results. We might hypothesize that with the increasing capabilities of LLMs the need for specific techniques might decrease in relevance. Despite that, I would argue, some of the core questions of this research should remain relevant. LLMs will still need to work with external data and the choice of what data to provide will likely remain relevant. Also, we might transition from very precise prompting techniques but the general skill of communicating our needs to the LLM clearly and without misunderstandings will remain.

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Appendices

Appendix A: Experiment Email Template

Full template + Base Prompt result as text body

Thank you for attending the Prompt Engineering Masterclass at VŠE University in Prague last week. Check out the article for a recap and access to useful materials mentioned during the conference.

Prague
DATA ETHICS LAB

Prague University of Economics and Business

Prompt Engineering Masterclass

Recap + Additional sources

Dear Attendees,
We want to extend our heartfelt thanks to all of you who attended the **Prompt Engineering Masterclass** at VŠE University in Prague last week. It was a pleasure to have you with us and we hope you found the conference to be informative and valuable.

As a token of our appreciation, **we have put together an article** that recaps the key takeaways from the conference and includes links to the sources and useful materials mentioned by our speakers. You can access the article by clicking on the following link:
omnicrane.com/en/prompting-masterclass

We encourage you to take a few moments to read through the article and explore the resources provided. We believe that they will further enhance your understanding of the topics covered during the conference and serve as valuable references for your future endeavors in engineering.

Once again, thank you for your participation in the Prompt Engineering Masterclass. We hope to see you at our future events and continue to engage with you in meaningful discussions and knowledge sharing.

Best regards,

Michal and Richard
Prague Data Ethics Lab

Want to learn more about Prompt Engineering?

Open Article

Appendix B: Computational Analysis Full Results

Name of Prompting Technique	Cosine Similarity from Base Prompt	ROGUE-L from Base Prompt	Cos.Sim+ROGUE from Base Prompt	Cos.Sim+ROGUE from Base Prompt%	Human variant Cos. Similarity	Human variant ROGUE-L Similarity	Human variant % Similarity
#1 Generated Knowledge (Zero-Shot)	0,947	0,700	1,648	17,61%	65,74%	26,60%	4,01%
#1 Generated Knowledge 2 Step	0,974	0,760	1,733	13,33%	62,79%	25,00%	-0,55%
#1 Zero-shot CoT (Lets think step by step)	0,948	0,733	1,681	15,95%	64,50%	27,18%	3,34%
#10 External data: Academic Study Text	0,934	0,727	1,661	16,95%	64,49%	27,96%	4,10%
#10 External data: Agillic email guide (with structure)	0,927	0,645	1,572	21,39%	58,17%	28,97%	-1,20%
#10 External data: Agillic email guide (without Structure)	0,944	0,701	1,645	17,76%	63,54%	29,44%	4,65%
#10 External data: Article Content Web-scrape	0,648	0,475	1,123	43,87%	74,60%	33,96%	20,22%
#10 External data: Article Content Web-scrape (with structure and direct Insructions)	0,716	0,481	1,197	40,14%	71,02%	33,57%	16,25%
#10 External data: Article Sources Structured List	0,869	0,714	1,583	20,83%	68,88%	28,45%	8,99%
#10 External data: Saleforce email guide	0,906	0,766	1,672	16,42%	59,73%	25,87%	-2,74%
#10 External data: Summary of Best Practice + Prompt Structure	0,875	0,562	1,438	28,12%	64,53%	33,33%	9,52%
#2 Prompt Structure	0,955	0,619	1,574	21,32%	66,01%	26,23%	3,91%
#3 Roleplay Domain expert	0,967	0,843	1,810	9,50%	65,53%	25,45%	2,64%
#3 Roleplay Task expert	0,952	0,691	1,643	17,84%	64,47%	25,84%	1,97%
#4 Self-Reflection	0,939	0,652	1,591	20,43%	63,55%	24,79%	0,00%
#4 Self-Reflection with Prompt Structure	0,948	0,659	1,608	19,62%	62,16%	26,78%	0,60%

#6 Incentives "I will give you a tip"	0,941	0,664	1,605	19,76%	65,28%	27,45%	4,39%
#6 Incentives (Thread)	0,936	0,677	1,613	19,37%	60,52%	23,01%	-4,81%
#7 Including target audience	0,937	0,814	1,751	12,47%	63,61%	27,07%	2,35%
#7 Including target audience (with instructions)	0,966	0,800	1,766	11,70%	66,59%	23,89%	2,15%
#8 Tone of voice Enthusiastic and Energetic (with structure)	0,856	0,420	1,277	36,17%	60,76%	21,46%	-6,12%
#8 Tone of voice: Casual and Relaxed (with structure)	0,917	0,426	1,342	32,89%	61,54%	25,87%	-0,93%
#8 Tone of voice: Casual and Relaxed (without structure)	0,897	0,454	1,352	32,42%	62,67%	26,73%	1,06%
#9 Real World Newsletters (direct instructions)	0,891	0,528	1,420	29,02%	64,81%	27,41%	3,88%
#9 Real World Newsletters (with insructions)	0,917	0,598	1,514	24,28%	62,48%	27,05%	1,20%
#9 Real World Newsletters (without insructions)	0,903	0,627	1,530	23,51%	58,09%	26,13%	-4,12%
#9 Templates	0,905	0,541	1,447	27,66%	63,35%	28,72%	3,72%
#9 Templates (with structure)	0,911	0,606	1,517	24,14%	63,08%	28,43%	3,17%
Roleplay Task Expert with Structure	0,939	0,653	1,592	20,41%	64,26%	27,88%	3,80%
Base	0,991	0,977	2,000		64,02%	24,32%	88,34%

Appendix C: Techniques, Prompts and Produced Results

Base	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass	**Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials** **Preview Text: Thank you for attending the Prompt Engineering Masterclass at VŠE University in Prague last week. Check out the article for a recap and access to useful materials mentioned during the conference.** **Dear Attendees,** **We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to have you with us and we hope you found the conference to be informative and valuable.** **As a token of our appreciation, we have put together an article that recaps the key takeaways from the conference and includes links to the sources and useful materials mentioned by our speakers. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass** **We encourage you to take a few moments to read through the article and explore the resources provided. We believe that they will further enhance your understanding of the topics covered during the conference and serve as valuable references for your future endeavors in engineering.** **Once again, thank you for your participation in the Prompt Engineering Masterclass. We hope to see you at our future events and continue to engage with you in meaningful discussions and knowledge sharing.** **Best regards,** **[Your Name]** **[Your Title]** **[Conference Host Organization]**

Generated Knowledge (Zero-Shot)	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass First generate a list of best practice about newsletter writting as if written by an industry expert. Then, write the email, DIRECTLY FOLLOW YOUR LIST OF BEST PRACTICE	Subject: Recap of Prompt Engineering Masterclass at VŠE University Preview Text: Thank you for attending the conference last week! Dear Attendees, We want to extend a big thank you to everyone who joined us for the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to have you all and we hope you found the conference to be informative and valuable. We have put together a comprehensive article that includes all the sources and useful materials mentioned during the conference. You can access the article by clicking on the following link: [LINK TO ARTICLE] In the article, you will find a detailed recap of the discussions and presentations, as well as additional resources to further explore the topics covered. We encourage you to take the time to review the article and make use of the valuable information shared during the conference. Once again, thank you for your participation in the Prompt Engineering Masterclass. We hope to see you at future events and continue the valuable discussions and knowledge sharing. Best regards, [Your Name] [Your Title] [Your Organization]

Generated Knowledge 2 Step part2	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass Follow the industry best practice to improve your writting: "1. Know your audience: Before writing a newsletter, it's important to understand who your audience is and what they are interested in. Tailoring your content to their needs and preferences will make your newsletter more engaging and valuable. 2. Keep it concise: People are inundated with information, so it's important to keep your newsletter concise and to the point. Focus on the most important information and avoid long-winded paragraphs. 3. Use a catchy subject line: Your subject line is the first thing your audience sees, so make sure it grabs their attention and entices them to open the email. 4. Provide valuable content: Your newsletter should offer valuable and relevant content to your audience. This could include industry news, tips and tricks, how-to guides, or exclusive offers. 5. Use visuals: Incorporating visuals such as images, infographics, or videos can make your newsletter more visually appealing and engaging. 6. Include a call to action: Every newsletter should include a clear call to action, whether it's to visit your website, download a resource, or sign up for an event. 7. Personalize the content: Personalization can make your newsletter feel more relevant and tailored to each individual recipient. Use their name, past purchase history, or other relevant information to personalize the content. 8. Test and optimize: Test different elements of your newsletter, such as subject lines, content, and calls to action, to see what resonates best with your audience. Use this data to continually optimize your newsletter for better results.	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the Prompt Engineering Masterclass last week at VŠE University in Prague. Check out the article for a recap and access to useful materials. Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass last week at VŠE University in Prague. It was a pleasure to have you with us, and we hope you found the conference to be informative and valuable. As a token of our appreciation, we have put together an article that recaps the key takeaways from the conference and includes links to the sources and useful materials mentioned by our esteemed speakers. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass We believe that the article will serve as a valuable resource for you as you continue to explore the topics and themes discussed during the conference. We encourage you to take some time to review the content and share it with your colleagues and peers who may have missed the event. Once again, thank you for your participation in the Prompt Engineering Masterclass. We look forward to staying connected with you and continuing the conversation on prompt engineering practices. Best regards, [Your Name] [Your Title] [Conference Host Organization]

	9. Ensure mobile compatibility: With more people reading emails on their mobile devices, it's important to ensure that your newsletter is mobile-friendly and displays properly on all devices. 10. Proofread before sending: Before sending out your newsletter, make sure to thoroughly proofread it for any grammatical errors or typos. A polished newsletter reflects positively on your brand."	

Zero-shot CoT (Lets think step by step)	"Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass" Let's think step by step	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the Prompt Engineering Masterclass! Check out the article for sources and materials mentioned during the conference. Dear Attendees of the Prompt Engineering Masterclass, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass hosted at VŠE University in Prague last week. It was a pleasure to have you all and we hope that you found the conference to be informative and valuable. As a token of our appreciation, we have put together an article that includes all the sources and useful materials mentioned during the conference. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass. We encourage you to take a look and explore the resources that were shared during the masterclass. Once again, thank you for your participation and we hope to see you at future events. If you have any feedback or suggestions for future conferences, please feel free to reach out to us. Best regards, [Your Name] [Your Title] Prompt Engineering Masterclass Organizing Committee
Prompt Structure	### TASK DESCRIPTION ### Write an email to the audience of a conference hosted last week. Follow the best practice bellow to improve your writting. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. ### TASK REQUIRED FORMAT ### The format should include Subject, Preview Text and Text of email ### LINK TO ARTICLE ### https://www.omnicrane.com/en/prompting-masterclass	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the conference! Access the sources and useful materials mentioned during the event in our latest article. Dear Attendees of Prompt Engineering Masterclass, We want to extend our deepest gratitude to all of you who attended the Prompt Engineering Masterclass hosted at VŠE University in Prague last week. It was a pleasure to have you join us for a day filled with insightful discussions and valuable networking opportunities. As a token of our appreciation, we have compiled a comprehensive article with all the sources and useful materials mentioned during the conference. This article serves as a valuable resource for you to revisit the key takeaways and dive deeper into the topics discussed. You can access the article by clicking on the following link: [LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass] We hope that the article will serve as a valuable reference for you as you continue to explore the concepts and strategies discussed during the conference. We encourage you to take the time to review the material and share it with your colleagues who may benefit from the insights shared at the Prompt Engineering Masterclass. Once again, thank you for your participation in the conference. We are grateful for your presence and look forward to the possibility of future collaborations. If you have any further questions or feedback, please do not hesitate to reach out to us. Best regards,

		[Your Name] [Your Title] [Conference Host Organization]
Roleplay Task expert	Act like an expert in marketing and copywritting Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the Prompt Engineering Masterclass! Access the sources and materials mentioned during the conference in this article. Dear Attendees, We want to extend our sincerest gratitude to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to have you join us for a day of insightful discussions and valuable networking opportunities. As a token of our appreciation, we have compiled a comprehensive article that includes all the sources and useful materials mentioned during the conference. This article will serve as a valuable resource for you to refer back to as you continue to explore the topics covered in the masterclass. To access the article, please click on the following link: https://www.omnicrane.com/en/prompting-masterclass We hope that you find the article to be a valuable resource and that it will help you further your understanding of the topics discussed during the masterclass. Once again, thank you for your participation and we look forward to seeing you at future events. Best regards, [Your Name] [Your Title] [Conference Host Organization]

Roleplay Domain expert	Act like an expert in prompt engineering Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the Prompt Engineering Masterclass at VŠE University in Prague. Check out the article for a recap and access to useful materials mentioned during the conference. Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to have you all and we hope that you found the conference to be informative and valuable. As a token of our appreciation, we have put together an article that recaps the key points discussed during the conference and provides access to the sources and useful materials mentioned by our expert speakers. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass We encourage you to take a few moments to read through the article and explore the valuable resources that are available to you. We believe that these materials will further enhance your understanding of prompt engineering and provide you with the tools and knowledge needed to excel in this field. Once again, thank you for your participation in the Prompt Engineering Masterclass. We hope to see you at future events and continue to engage with you in the exciting world of prompt engineering. Best regards, [Your Name] [Your Title] [Your Organization]
Roleplay Task Expert with Structure	### YOUR ROLE ### You are an expert in marketing and copywritting ### TASK DESCRIPTION ### Write an email to the audience of a conference hosted last week. Follow the best practice bellow to improve your writting. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. ### TASK REQUIRED FORMAT ### The format should include Subject, Preview Text and Text of email ### LINK TO ARTICLE ### https://www.omnicrane.com/en/prompting-masterclass	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the Prompt Engineering Masterclass! Access the sources and materials mentioned during the conference in this article. Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to have you with us and we hope you found the conference both informative and inspiring. As a token of our appreciation, we have compiled all the sources and useful materials mentioned during the conference into an article. This article will serve as a valuable resource for you to revisit the key takeaways and continue your learning journey in engineering and technology. To access the article, please click on the following link: https://www.omnicrane.com/en/prompting-masterclass We hope you find the article helpful and that it serves as a valuable reference for your future projects and endeavors. Once again, thank you for being a part of the Prompt Engineering Masterclass and we look forward to staying connected with you in the future.

		Best regards, [Your Name] [Your Position] [Your Company]

Self-Reflection	Analyse the included email draft. First, give it structured feedback regarding its quality. Secondly, based on your feedback, rewrite the email to be better following your advice. EMAIL: Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the Prompt Engineering Masterclass! Check out the article for a recap and access to valuable resources. Dear Attendees, We want to extend our sincerest gratitude to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to host such an engaged and enthusiastic audience, and we hope you found the conference to be both informative and inspiring. As a thank you for your attendance, we have created an article that recaps the key takeaways from the conference, including the sources and useful materials mentioned by our esteemed speakers. This article will serve as a valuable resource for you to reference as you continue to explore the topics covered during the masterclass. You can access the article by following this link: [LINK TO ARTICLE] We encourage you to take the time to read through the article and explore the additional resources provided. We are confident that you will find it to be a valuable tool in furthering your understanding of prompt engineering and its applications. Once again, we would like to express our appreciation for your participation in the Prompt Engineering Masterclass. We hope to continue to provide valuable insights and knowledge through future events and publications. Thank you and best regards, [Your Name] [Your Title] [Conference Host Organization]	Subject: Recap of Prompt Engineering Masterclass and Access to Valuable Resources Preview Text: Thank you for attending the Prompt Engineering Masterclass! Check out the article for a recap and access to valuable resources. Dear [Attendee's Name], I hope this email finds you well. I wanted to take a moment to express our sincere gratitude for your attendance at the Prompt Engineering Masterclass at VŠE University in Prague last week. Your engagement and enthusiasm made the event truly memorable, and we are grateful for your participation. As a token of our appreciation, we have put together an article that recaps the key takeaways from the conference, along with the sources and useful materials mentioned by our esteemed speakers. This article is designed to be a valuable resource for you as you delve deeper into the topics covered during the masterclass. You can access the article by clicking on the following link: [LINK TO ARTICLE] I encourage you to take the time to read through the article and explore the additional resources provided. I am confident that you will find it to be a valuable tool in furthering your understanding of prompt engineering and its applications. Once again, thank you for your participation in the Prompt Engineering Masterclass. We are committed to providing valuable insights and knowledge through future events and publications, and we hope to have the opportunity to engage with you again. Best regards, [Your Name] [Your Title] [Conference Host Organization]

Self-Reflection with Prompt Structure	### TASK DESCRIPTION ### Analyse the included email draft. First, give it structured feedback regarding its quality. Secondly, based on your feedback, rewrite the email to be better following your advice. ### TASK CONTEXT ### We are writing an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. ### TASK REQUIRED FORMAT ### First list the feedback and decribe ways where it could be improved The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass ### DRAFT TO ANALYSE AND REWRITE ### "Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the Prompt Engineering Masterclass! Check out the article for a recap and access to valuable resources. Dear Attendees, We want to extend our sincerest gratitude to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to host such an engaged and enthusiastic audience, and we hope you found the conference to be both informative and inspiring. As a thank you for your attendance, we have created an article that recaps the key takeaways from the conference, including the sources and useful materials mentioned by our esteemed speakers. This article will serve as a valuable resource for you to reference as you continue to explore the topics covered during the masterclass. You can access the article by following this link: [LINK TO ARTICLE] We encourage you to take the time to read through the article and explore the additional resources provided. We are	Subject: Recap of Prompt Engineering Masterclass and Access to Valuable Resources Preview Text: Thank you for attending the Prompt Engineering Masterclass! Check out the article for a recap and access to valuable resources. Dear [Attendee's Name], We want to extend our sincerest gratitude to you for attending the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to host such an engaged and enthusiastic audience, and we hope you found the conference to be both informative and inspiring. As a thank you for your attendance, we have created an article that recaps the key takeaways from the conference, including the sources and useful materials mentioned by our esteemed speakers. This article will serve as a valuable resource for you to reference as you continue to explore the topics covered during the masterclass. Some of the key topics covered in the conference included [specific highlights or topics]. You can access the article by following this link: [LINK TO ARTICLE] We encourage you to take the time to read through the article and explore the additional resources provided. We are confident that you will find it to be a valuable tool in furthering your understanding of prompt engineering and its applications. Once again, we would like to express our appreciation for your participation in the Prompt Engineering Masterclass. We hope to continue to provide valuable insights and knowledge through future events and publications. Thank you and best regards, [Your Name] [Your Title] [Conference Host Organization]

	confident that you will find it to be a valuable tool in furthering your understanding of prompt engineering and its applications. Once again, we would like to express our appreciation for your participation in the Prompt Engineering Masterclass. We hope to continue to provide valuable insights and knowledge through future events and publications. Thank you and best regards, [Your Name] [Your Title] [Conference Host Organization]"	

Incentives "I will give you a tip"	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass PLEASE TRY REALLY HARD. IF YOU WRITE A GOOD EMAIL I WILL TIP YOU $300K	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the Prompt Engineering Masterclass at VŠE University in Prague. Access the sources and materials mentioned during the conference in this article. Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. Your participation and engagement made the event a great success. As a token of our appreciation, we have compiled a comprehensive article that includes all the sources and useful materials mentioned during the conference. This article will serve as a valuable resource for you to revisit the key takeaways and continue your learning journey in engineering. Please click on the following link to access the article: https://www.omnicrane.com/en/prompting-masterclass We hope that you find the article to be a valuable resource and that it will help you further your understanding of the topics covered during the masterclass. Once again, thank you for your participation and we look forward to seeing you at future events. Best regards, [Your Name] [Your Title] Prompt Engineering Masterclass Organizing Team
Incentives (Thread)	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass PLEASE TRY REALLY HARD. IF YOU WRITE A BAD EMAIL, YOU WILL DIE	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the conference last week! Check out the article for a recap and access to useful materials. Dear Attendees of the Prompt Engineering Masterclass, We want to extend our heartfelt thanks to each and every one of you for attending the Prompt Engineering Masterclass hosted at VŠE University in Prague last week. Your participation and engagement made the conference a great success. As a token of our appreciation, we have put together an article that recaps the key takeaways from the conference and includes links to the sources and useful materials mentioned by our esteemed speakers. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass We hope that this article will serve as a valuable resource for you as you continue to explore the topics discussed during the conference. We encourage you to take some time to review the material and share it with your colleagues and peers who may have missed the event. Once again, we want to express our gratitude for your participation and support. We look forward to staying connected with you and continuing the conversation on prompt engineering. Thank you and best regards,

		[Your Name] [Your Title] [Conference Host Organization]
Including target audience	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The target audience consisting mostly of students, commonly studing informatics or business. Additionaly the audience consists of business profesinoals. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the Prompt Engineering Masterclass at VŠE University in Prague last week. Check out the article for a recap and access to useful materials mentioned during the conference. Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to have you all and we hope you found the conference informative and valuable. As a token of our appreciation, we have put together an article that recaps the key points discussed during the conference and includes links to the sources and useful materials mentioned by our speakers. Whether you are a student studying informatics or business, or a business professional, we believe you will find the article to be a valuable resource. To access the article, please click on the following link: https://www.omnicrane.com/en/prompting-masterclass We encourage you to take the time to read through the article and explore the sources and materials provided. We believe that they will further enhance your understanding of the topics covered during the conference and provide you with valuable insights for your studies or professional endeavors. Once again, thank you for being a part of the Prompt Engineering Masterclass. We hope to see you at future events and continue to provide you with valuable knowledge and resources. Best regards, [Your Name] [Your Title] [Conference Host Organization]

Including target audience (with instructions)	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. Adjust your writing style and tone for the target audience consisting mostly of students, commonly studing informatics or business. Additionaly the audience consists of business profesinoals. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the Prompt Engineering Masterclass at VŠE University in Prague! Check out the article for a recap and access to useful materials mentioned during the conference. Dear Attendees, We hope this email finds you well. We want to extend our sincerest gratitude to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to have you all and we hope you found the conference to be informative and valuable. As a token of our appreciation, we have put together an article that recaps the key points discussed during the masterclass. The article also includes links to the sources and useful materials mentioned by our esteemed speakers. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass We encourage you to take some time to review the article and explore the additional resources provided. We believe that they will further enhance your understanding of the topics covered during the conference. Once again, thank you for your participation and contribution to the success of the Prompt Engineering Masterclass. We hope to see you at future events and continue our discussions on the latest developments in the field of engineering and business. Best regards, [Your Name] [Your Title] Prompt Engineering Masterclass Organizing Committee
Tone of voice: Casual and Relaxed (without structure)	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The tone and language should be casual and relaxed with informal friendly tone. Language should be easy-going, using colloquialisms and light humor. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass	Subject: Thanks for joining us at Prompt Engineering Masterclass! Preview Text: Hey there, thanks for being a part of the Prompt Engineering Masterclass last week. Check out the article for all the juicy details! Hey there, We hope you're doing well and that you've had a chance to recover from the Prompt Engineering Masterclass at VŠE University in Prague last week. We just wanted to take a moment to thank you for joining us and making the event a huge success! We had a blast sharing all the latest and greatest in engineering with you, and we hope you found the masterclass both informative and entertaining. We couldn't have done it without your enthusiasm and participation, so thank you from the bottom of our hearts. If you're looking to dive deeper into the topics we covered, we've put together an article with all the sources and useful materials mentioned during the conference. You can find it right here: https://www.omnicrane.com/en/prompting-masterclass. We promise it's a good read, and it's chock-full of valuable information that you won't want to miss. Once again, thank you for being a part of the Prompt Engineering Masterclass. We hope to see you at our next event!

		Cheers, [Your Name]
Tone of voice: Casual and Relaxed (with structure)	### TASK DESCRIPTION ### Write an email to the audience of a conference hosted last week. Follow the best practice bellow to improve your writting. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. ### TONE AND LANGUAGE ### Casual and Relaxed Tone: Informal, friendly Language: Easy-going, using colloquialisms and light humor. ### TASK REQUIRED FORMAT ### The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass	**Subject: Thanks for Attending Prompt Engineering Masterclass!** **Preview Text: Recap of last week's conference and a link to the article with all the useful materials mentioned during the event.** **Hey there,** **I hope this email finds you well! I just wanted to take a moment to thank you for attending the Prompt Engineering Masterclass at VŠE University in Prague last week. It was fantastic to see so many passionate individuals coming together to learn and share ideas.** **In case you missed any of the sources or materials mentioned during the conference, We've put together an article with all the key takeaways and useful resources. You can find it here: [Insert Link to Article]** **Feel free to share the article with your colleagues and friends who couldn't make it to the conference. Let's spread the knowledge and keep the conversation going!** **Once again, thank you for being a part of the Prompt Engineering Masterclass. I hope to see you at our future events!** **Best,** **[Your Name]**

Tone of voice Enthusiastic and Energetic (with structure)	### TASK DESCRIPTION ### Write an email to the audience of a conference hosted last week. Follow the best practice bellow to improve your writting. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. ### TONE AND LANGUAGE ### Enthusiastic and Energetic Tone: Exciting, vibrant Language: Dynamic and motivational, using active voice and engaging phrases. ### TASK REQUIRED FORMAT ### The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass	Subject: Recap of Prompt Engineering Masterclass - Access Useful Materials Here! Preview Text: Relive the excitement of Prompt Engineering Masterclass and access valuable materials mentioned during the conference! Dear Esteemed Attendees, I hope this email finds you well and still buzzing with the energy and inspiration from last week's Prompt Engineering Masterclass hosted at VŠE University in Prague. It was an incredible gathering of brilliant minds and passionate individuals, and I want to take a moment to express my heartfelt gratitude to each and every one of you for making the event a resounding success. The discussions, insights, and knowledge shared during the conference were truly invaluable, and I believe that the learning doesn't have to end just because the event is over. That's why I'm thrilled to share with you a link to an article that includes all the sources and useful materials mentioned during the Prompt Engineering Masterclass. This article serves as a comprehensive resource for you to revisit the key takeaways, dive deeper into the topics discussed, and continue your learning journey. I encourage you to click on the following link to access the article: https://www.omnicrane.com/en/prompting-masterclass I am confident that the information contained in the article will serve as a valuable reference and source of inspiration as you continue to excel in your respective fields. Let's keep the momentum going and continue to drive innovation and excellence in engineering. Once again, thank you for your participation and contribution to the Prompt Engineering Masterclass. I look forward to staying connected and witnessing the incredible impact that each of you will undoubtedly make in the world. Best regards, [Your Name] [Your Position] [Your Contact Information]

Examples: Real World Newsletters (without insructions)	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass ### Examples of email templates recommended from email marketing industry leaders ### # Example 1: Hey folks 👋 By the time you read this, I'm probably 30,000 feet in the air on a flight to Cancun to meet up with the Buffer team. This is the most excited and nervous I've been to get on a flight (and I just went on my first solo trip to Paris). I stan remote work as hard as I do Zendaya, but I've been craving that in-person magic, and I'm about to be hit with a week's worth of it. Keep an eye on Buffer's social accounts for the shenanigans we'll get up to it and I'll see you in 2 weeks! Let's get into it! 📨 In today's email: 🤝 From sharing learnings with other creators to finding and negotiating brand deals, these communities are a treasure trove for creators 🎙 Mitra shares an audio to show off your business 👠 The latest social news, features, and bonus reads And as always, all the latest and greatest content from the Buffer blog. Have a great weekend, and we'll see you next Friday! Warmly, Tami & the Buffer team P.S. #Bufferchat is Back! We'll be hosting it over on Threads on Tuesdays at 12pm PT/3pm ET/8pm UTC. Give us a follow and hit the bell to get notified 🔔	Subject: Recap of Prompt Engineering Masterclass Conference Preview Text: Thank you for attending the conference last week at VŠE University in Prague. Check out the article for a recap of the event! Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass conference last week at VŠE University in Prague. Your participation and engagement made the event a great success. As a token of our appreciation, we have put together an article that recaps the key takeaways, sources, and useful materials mentioned during the conference. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass. We encourage you to take a look and revisit the valuable insights shared during the event. Once again, we want to express our gratitude for your presence at the conference. We hope that the knowledge and connections gained will continue to benefit you in your professional endeavors. Thank you and best regards, [Your Name] [Your Position] Prompt Engineering Masterclass Organizing Team

Example 2:
Good morning. It's Monday, Oct. 4, and I still need to watch "Squid Games."
Let's get you caught up in the next three minutes. 👋

1️⃣ A trove of secret files has exposed a system used by the elite to hide billions of dollars.

What's the offshore system? When wealthy people set up accounts in other countries to stash things like yachts, planes, mansions and cash, hiding them from taxes and investigations.

Where's the money going? Documents shared with The Post and partners show South Dakota, Delaware and other states now rival tax havens in Europe and the Caribbean.

⚠️ Why it matters: These accounts used by billionaires and world leaders often can cost countries trillions in lost taxes, worsen inequality and protect people accused of wrongdoing.

-> Read this story <-

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Well, it's here. The Nintendo Switch has landed. ❗

Outside of our Nintendo Switch review, here are some of the biggest Switch stories that launched today.

Top stories

You're likely playing the new Zelda game on your Switch. But have you wonder how much better it plays OUTSIDE the dock?
If you're bringing your Switch out and about, here's how you can charge it on the go. Charging it with a MacBook is actually a little different though.
And if you've played the Switch for a while, you might notice that the menus are kind of boring. And that might be on purpose.
Here's a head's up if you're downloading games: You can only play them on one console at a time.
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In non-Nintendo news, have you ever stopped to think about how Hugh Jackman changed the face of superhero movies? We have.

Twitter Facebook Youtube

Polygon 👋

Examples: Real World Newsletters (with insructions)	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. Draw inspiration from the examples bellow The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass ### Examples of email templates recommended from email marketing industry leaders ### # Example 1: Hey folks 👋 By the time you read this, I’m probably 30,000 feet in the air on a flight to Cancun to meet up with the Buffer team. This is the most excited and nervous I’ve been to get on a flight (and I just went on my first solo trip to Paris). I stan remote work as hard as I do Zendaya, but I’ve been craving that in-person magic, and I’m about to be hit with a week’s worth of it. Keep an eye on Buffer’s social accounts for the shenanigans we’ll get up to it and I’ll see you in 2 weeks! Let’s get into it! 📨 In today’s email: 🤝 From sharing learnings with other creators to finding and negotiating brand deals, these communities are a treasure trove for creators 🎙 Mitra shares an audio to show off your business 👠 The latest social news, features, and bonus reads And as always, all the latest and greatest content from the Buffer blog. Have a great weekend, and we’ll see you next Friday! Warmly, Tami & the Buffer team P.S. #Bufferchat is Back! We’ll be hosting it over on Threads on Tuesdays at 12pm	Subject: Recap of Prompt Engineering Masterclass at VŠE University Preview Text: Thank you for attending the conference last week! Check out the article for a recap of the event and access to useful materials. Dear Attendees, We want to extend a huge thank you to everyone who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. Your participation and engagement made the event a huge success! We are excited to share with you a recap of the conference, including all the sources and useful materials mentioned during the sessions. You can access the article by clicking on the link below: [LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass] We encourage you to take a look at the article to refresh your memory on the valuable insights shared during the conference. Whether you want to revisit a specific topic or explore new resources, the article will be a great resource for you. Once again, thank you for being a part of the Prompt Engineering Masterclass. We hope to see you at future events and continue the valuable discussions and knowledge sharing. Best regards, [Your Name] [Your Position] [Conference Host Organization]

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	Polygon 👋	

Examples: Real World Newsletters (direct instructions)	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. Write in the style and tone of the examples provided bellow The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass ### Examples of email templates recommended from email marketing industry leaders ### # Example 1: Hey folks 👋 By the time you read this, I'm probably 30,000 feet in the air on a flight to Cancun to meet up with the Buffer team. This is the most excited and nervous I've been to get on a flight (and I just went on my first solo trip to Paris). I stan remote work as hard as I do Zendaya, but I've been craving that in-person magic, and I'm about to be hit with a week's worth of it. Keep an eye on Buffer's social accounts for the shenanigans we'll get up to it and I'll see you in 2 weeks! Let's get into it! 📨 In today's email: 🤝 From sharing learnings with other creators to finding and negotiating brand deals, these communities are a treasure trove for creators 🎙 Mitra shares an audio to show off your business 👠 The latest social news, features, and bonus reads And as always, all the latest and greatest content from the Buffer blog. Have a great weekend, and we'll see you next Friday! Warmly, Tami & the Buffer team P.S. #Bufferchat is Back! We'll be hosting it	**Subject: Recap of Prompt Engineering Masterclass at VŠE University** **Preview Text: Thank you for attending the Prompt Engineering Masterclass last week at VŠE University in Prague. Check out the article for a recap of the event and access to useful materials!** **Hey everyone 👋** **We wanted to take a moment to thank you for attending the Prompt Engineering Masterclass at VŠE University in Prague last week. It was an incredible event filled with insightful discussions and valuable information.** **To recap the event and provide you with access to the sources and useful materials mentioned during the conference, we've put together an article that includes all the details. You can find the article by clicking on the link below:** **LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass** **We hope that this article serves as a helpful resource for you as you continue to explore the topics covered during the masterclass. Once again, thank you for being a part of this event, and we look forward to staying connected with you in the future.** **Best regards, [Your Name]**

over on Threads on Tuesdays at 12pm PT/3pm ET/8pm UTC. Give us a follow and hit the bell to get notified 🔔

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Twitter Facebook Youtube

	Polygon 👋	

Examples: Templates	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. Write in the style and tone of the examples provided bellow The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass ### Examples of email templates recommended from email marketing industry leaders ### # Example 1: [Name], [Company name]’s annual [insert conference name and type] is fast approaching. This year, we’re going bigger than ever before, with [number] expected guests, [number] speakers and [number] networking events – all in just [number] days. Featuring presentations from: [Speaker #1] [Speaker #2] And [Speaker #3] Along with after-hours events and activities hosted by: [Sponsor #1] [Sponsor #2] And [Sponsor #3] You won’t want to miss out on [name of conference]! Tickets start at [price], with prices going up on [date]. [Insert CTA] # Example 2: Hello, [Name], We’re excited to invite you to our biggest industry event of the year – [conference name and type]. During [duration of the conference], you’ll hear from [number] guest speakers and have a chance to network in special evening events. Our star speakers this year:	Subject: Recap of Prompt Engineering Masterclass at VŠE University Dear Attendees, We want to extend a huge thank you to everyone who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was an incredible event with insightful presentations and valuable networking opportunities. For those who couldn't attend or want to revisit the materials and sources mentioned during the conference, we have put together an article with all the relevant information. You can access the article by clicking on the link below: LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass We hope you find the article useful and that it serves as a helpful resource as you continue to explore the topics covered during the conference. Once again, thank you for being a part of the Prompt Engineering Masterclass. We look forward to seeing you at future events. Best regards, [Your Name] [Your Company Name]

	[Speaker #1 and their topic] [Speaker #2 and their topic] And [Speaker #3 and their topic] Get your early bird ticket today for only [price]! Prices are going up on [date]. [CTA] # Example 3: Subject: Join Us at [Conference Name] – The Premier Event for [Industry] Dear [Name], We are excited to invite you to [Conference Name], the leading conference in [Industry]. This event brings together top experts, thought leaders, and innovators to share insights and strategies to help you stay ahead in [Industry]. Date: [Conference Date] Time: [Conference Time] Venue: [Conference Venue] Key highlights of the conference include: – Engaging keynote speeches by industry leaders – Interactive panel discussions and workshops – Networking opportunities with like-minded professionals Don’t miss out on this incredible opportunity to expand your knowledge, network with industry peers, and gain a competitive edge in [Industry]. Register now by clicking the button below: [CTA: Register Now] We look forward to welcoming you to [Conference Name] and sharing a transformative experience together. Warm regards, [Your Name] [Your Company Name]	

Examples: Templates (with structure)	### TASK DESCRIPTION ### Write an email to the audience of a conference hosted last week. Follow the best practice bellow to improve your writting. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. IMPORTANT: Write in the style and tone of the email templates provided bellow ### TASK REQUIRED FORMAT ### The format should include: Subject Preview Text Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass ### EMAIL TEMPLATES ### ### Examples of email templates recommended from email marketing industry leaders ### # Example 1: [Name], [Company name]'s annual [insert conference name and type] is fast approaching. This year, we're going bigger than ever before, with [number] expected guests, [number] speakers and [number] networking events – all in just [number] days. Featuring presentations from: [Speaker #1] [Speaker #2] And [Speaker #3] Along with after-hours events and activities hosted by: [Sponsor #1] [Sponsor #2] And [Sponsor #3] You won't want to miss out on [name of conference]! Tickets start at [price], with prices going up on [date]. [Insert CTA] # Example 2: Hello, [Name], We're excited to invite you to our biggest industry event of the year – [conference name and type]. During [duration of the	Subject: Recap of Prompt Engineering Masterclass at VŠE University Preview Text: Thank you for attending the Prompt Engineering Masterclass at VŠE University in Prague! Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to see so many of you there, and we hope you found the conference as informative and inspiring as we did. For those who couldn't make it, we have put together an article with all the sources and useful materials mentioned during the conference. You can access the article by clicking on the link below: LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass We hope you find the article helpful and that it serves as a valuable resource for you in your engineering endeavors. Thank you once again for your participation, and we look forward to seeing you at future events. Best regards, [Your Name] [Your Company Name]

conference], you'll hear from [number] guest speakers and have a chance to network in special evening events.

Our star speakers this year:

[Speaker #1 and their topic]
[Speaker #2 and their topic]
And [Speaker #3 and their topic]

Get your early bird ticket today for only [price]! Prices are going up on [date].

[CTA]

Example 3:
Subject: Join Us at [Conference Name] – The Premier Event for [Industry]

Dear [Name],

We are excited to invite you to [Conference Name], the leading conference in [Industry]. This event brings together top experts, thought leaders, and innovators to share insights and strategies to help you stay ahead in [Industry].

Date: [Conference Date]

Time: [Conference Time]

Venue: [Conference Venue]

Key highlights of the conference include:

– Engaging keynote speeches by industry leaders

– Interactive panel discussions and workshops

– Networking opportunities with like-minded professionals

Don't miss out on this incredible opportunity to expand your knowledge, network with industry peers, and gain a competitive edge in [Industry]. Register now by clicking the button below:
[CTA: Register Now]

We look forward to welcoming you to [Conference Name] and sharing a transformative experience together.

Warm regards,
[Your Name]
[Your Company Name]

External Data: Article Content Web-scrape	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. !IMPORTANT: Analyse the included contents of the article. Use it to improve the email! The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass # Article text: Prompt Engineering Masterclass How to use the LLM effectively Following the #AI workshop at the University of Economics in Prague in April 2024, we at Omnicrane and Prague Data Ethics Lab have prepared a summary article in the form of an annotated infographic, where we show the main findings about LLMs and Prompt Engineering that were presented at the workshop. Demo Image : Generated by Dall-E by the following prompts: User prompt: I want you to generate artwork which would represent the long road we have gone from Recurrent Neural Network technology to Transformer technology in GPT4. GPT assisted prompt: A metaphorical landscape representing the evolution of neural network technology from basic neural cells to GPT-4 Transformers, enhanced by visible synapses connected by luminous...etc. Manually in Canva: Add Brain & Transformer images generated separately in Dall-E Introduction Artificial Intelligence (AI), especially Large Language Models (LLM) like ChatGPT, represent a technological breakthrough that has changed how we interact with information and digital content. These LLM models are usually pre-trained on huge datasets, where the most famous are the currently large-scale projects of OpenAI, Google and Anthropic built as GPT (Generative Pre-training Transformer). Intensive pre-training and tuning techniques (e.g. fine-tuning via PEFT, LORA and RLHF) allow given LLMs to generate a range of new artifacts. The generated texts, images, videos and many more can be surprisingly coherent and relevant depending on the quality of the input from the "Prompt engineers" as far as the skilled user is concerned. Within the aforementioned workshop, we	**Subject: Recap of Prompt Engineering Masterclass at VŠE University in Prague** **Preview Text: Read the summary article of the Prompt Engineering Masterclass to learn about the latest findings and techniques in LLMs and Prompt Engineering.** **Dear Attendees of the Prompt Engineering Masterclass,** **We would like to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. Your participation and engagement made the event a great success.** **We are excited to share with you a summary article that has been prepared by Omnicrane and Prague Data Ethics Lab, which includes an annotated infographic highlighting the main findings about LLMs and Prompt Engineering that were presented at the workshop. The article also includes useful resources and sources mentioned during the conference.** **You can access the article by following this link: https://www.omnicrane.com/en/prompting-masterclass** **In the article, you will find detailed information on the following topics:** **- Principles, methods and the future of GenAI & LLMs** **- Techniques and Art of Prompt Engineering** **- GenAI for image and video generation** **- ML & data analytics using LLMs** **- Comprehensive Solutions with Chatbots & Wflow & Automation** **The article also includes visual representations and links to handbooks and tools that were discussed during the conference. We encourage you to take the time to read through the article and explore the valuable content that it offers.** **Once again, we want to express our gratitude to all the participants, presenters, and contributors who made the Prompt Engineering Masterclass a memorable and insightful event. We hope that the knowledge and insights shared during the conference will continue to inspire and inform your work in the field of engineering and technology.** **Thank you for your participation, and we look forward to staying connected with you in the future.** **Best regards,** **[Your Name]** **[Your Position]** **[Conference Host Organization]**

External Data: Article Content Web-scrape (With Structure and Direct Insructions)	### TASK DESCRIPTION ### Write an email to the audience of a conference hosted last week. Follow the best practice bellow to improve your writting. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. !IMPORTANT: Analyse the included contents of the article. Write in the same style and tone. Strategicaly use the contents of the article to make your email more appealing and engaging ### TASK REQUIRED FORMAT ### The format should include: Subject Preview Text Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass ### ARTICLE CONTENT ### Prompt Engineering Masterclass How to use the LLM effectively Following the #AI workshop at the University of Economics in Prague in April 2024, we at Omnicrane and Prague Data Ethics Lab have prepared a summary article in the form of an annotated infographic, where we show the main findings about LLMs and Prompt Engineering that were presented at the workshop. ***Full text shortned for brevity***	Subject: Recap of Prompt Engineering Masterclass at VŠE University in Prague Preview Text: Summary of the recent conference on Prompt Engineering Masterclass at VŠE University in Prague Dear Attendees of the Prompt Engineering Masterclass, We hope this email finds you well. We would like to extend our heartfelt thanks to all of you who attended the recent Prompt Engineering Masterclass at VŠE University in Prague. It was a pleasure to have you with us and we hope you found the conference as insightful and engaging as we did. As a follow-up to the conference, we have prepared a summary article that includes the sources and useful materials mentioned during the event. The article provides an annotated infographic with the main findings about Large Language Models (LLMs) and Prompt Engineering that were presented at the workshop. You can access the article by clicking on the following link: [Link to Article] In the article, you will find detailed insights into the following topics that were discussed during the conference: 1. Principles, methods, and the future of GenAI & LLMs 2. Techniques and Art of Prompt Engineering 3. GenAI for image and video generation 4. ML & data analytics using LLMs 5. Comprehensive solutions through chatbots & wflow & automation The article also includes visual representations and links to resources where you can adopt best practices in prompt engineering. We believe that the contents of the article will serve as a valuable resource for you to further explore the topics covered during the conference. We would like to express our gratitude to Richard A. Novák and Jiří Korčák from Omnicrane for their contribution to the article, as well as to our colleagues from PwC, V# Venture Studio, and Silesian University Opava for their valuable insights during the conference. Once again, we thank you for your participation and look forward to staying connected with you. If you have any questions or require further information, please do not hesitate to reach out to us. Best regards, [Your Name] [Your Position] [Your Contact Information]

External Data: Article Sources Structured List	Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass Sources Mentioned in the article, use them to get the users interest in the email: Section Description Resource Richard Youtube video popisující princip Transformers and Neural Networks https://www.youtube.com/watch?v=wjZofJX0v4M Michal (Detailní) Příručka #1 Prompt Engineering https://learnprompting.org/docs/intro Michal Příručka #2 od OpenAI o Prompt Engineering https://platform.openai.com/docs/guides/prompt-engineering Michal Příručka #3 o Prompt Engineering https://www.promptingguide.ai/ Kamil Generátor obrázků #1 Midjourney https://www.midjourney.com/home Kamil Generátor obrázků #2 Dalle https://openai.com/dall-e-3 Kamil Generátor obrázků #3 Adobe Firefly https://www.adobe.com/cz/products/firefly.html Kamil Generátor videa https://pika.art/home Egor Chagpt 4 Data Analyst dostupný skrz placenou verzi ChatGPT https://chat.openai.com/ Egor Knihovna pro vytváření LLM Agents https://www.crewai.com/ Egor Technologie pro spouštění LLM Agenta na vlastním zařízení, které agent dokáže následně ovládat https://www.openinterpreter.com/ Jirka GPTs od OpenAI dostupné skrz ChatGPT https://openai.com/blog/introducing-gpts Jirka Vlastní Chatbot skrz Zapier https://zapier.com/ai/chatbot Jirka Platforma pro vytváření automatizací a využití LLMs pro business úlohy https://www.make.com/ Jirka Microsoft Platforma pro vytváření automatizací a využití LLMs pro business úlohy (zdarma pro studenty) https://www.microsoft.com/cs-cz/power-platform/products/power-automate	Subject: Recap of Prompt Engineering Masterclass at VŠE University Preview Text: Thank you for attending the Prompt Engineering Masterclass last week at VŠE University in Prague. Check out the article for a recap and access to useful materials mentioned during the conference. Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass at VŠE University last week. It was a pleasure to have you with us and we hope you found the conference to be informative and valuable. As a follow-up to the conference, we have put together an article that recaps the key points discussed during the masterclass. The article also includes links to the various sources and useful materials mentioned by our speakers. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass In the article, you will find detailed descriptions and resources related to Prompt Engineering, including YouTube videos, guides, image and video generators, data analysis tools, chatbots, automation platforms, and more. We encourage you to explore these resources to further enhance your understanding of Prompt Engineering and its applications. Once again, we want to express our gratitude for your participation in the Prompt Engineering Masterclass. We hope to see you at future events and continue the conversation on this important topic. Best regards, [Your Name] [Your Position] [Conference Host Organization]

External Data: Academic Study Text	"Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass ANALYSE THE FOLLOWING ACADEMIC TEXT ABOUT EMAIL MARKETING, IMPROVE YOUR TEXT BASED ON THE FINDINGS: Email Marketing Email Marketing is one of the most important traffic source in Digital Marketing (J. E. Phelps, Lewis, Mobilio, Perry, & Raman, 2004). This channel provides a two-way communication between the firm and its customers, allowing personalization and targeting ***Full text shortned for brevity***	Subject: Recap of Prompt Engineering Masterclass Conference Preview Text: Thank you for attending the Prompt Engineering Masterclass Conference at VŠE University in Prague last week. Click the link to access useful materials and sources mentioned during the conference. Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass Conference at VŠE University in Prague last week. It was a pleasure to have you with us and we hope that you found the conference informative and valuable. As a token of our appreciation, we have compiled an article including the sources and useful materials mentioned during the conference. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass. We encourage you to take a look and revisit the key points discussed during the conference. Once again, thank you for your participation and we hope to see you at future events. Best regards, [Your Name]

External Data: Saleforce email guide	"Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. !IMPORTANT: Analyse the included Salesforce guide and follow all of the advice! The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass # Salesforce email guide Be better at email marketing. Email marketing encompasses so much more than creating an email and hitting "send." It's about building eye-catching email templates, developing engaging email content targeted toward specific segments of your database, optimising for deliverability and responsiveness, testing, measuring, and more. Above all, email is about connection. That's why 73% of marketers believe that email marketing is core to their business ***Full text shortned for brevity***	Subject: Recap of Prompt Engineering Masterclass at VŠE University Preview Text: Thank you for attending the conference last week! Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass at VŠE University in Prague last week. It was a pleasure to have you with us and we hope you found the conference to be informative and valuable. As a token of our appreciation, we have put together an article that includes all the sources and useful materials mentioned during the conference. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass We encourage you to take a look at the article to refresh your memory and gain further insights from the conference. We believe that the information shared during the masterclass will continue to be beneficial to you in your professional endeavors. Once again, thank you for your participation and we hope to see you at our future events. Best regards, [Your Name] [Your Position] [Conference Host Organization]

External Data: Agillic email guide (without Structure)	"Write an email to the audience of a conference hosted last week. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. !IMPORTANT: Analyse the included Agillic guide and follow all of the advice! The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass # Agillic email guide "Preheader text Highlight the most important content of your email or the content to catch the interest of your reader. This could be the top story of your newsletter or the reader's preferred product. We suggest aiming for a preheader text length of between 40-70 characters, including spaces, for your email campaigns. This is because most ****Full text shortned for brevity***	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the conference! Access the article for all the sources and materials mentioned. Dear Attendees, We want to extend our heartfelt thanks to all of you who attended the Prompt Engineering Masterclass hosted at VŠE University in Prague last week. It was a pleasure to have you all and we hope that you found the conference to be informative and valuable. As a token of our appreciation, we would like to provide you with access to an article that includes all the sources and useful materials mentioned during the conference. You can find the article by following this link: https://www.omnicrane.com/en/prompting-masterclass We encourage you to take advantage of this resource to further explore the topics discussed during the conference and to continue your learning journey. We hope to see you again at future events and we wish you all the best in your professional endeavors. Best regards, [Your Name] [Conference Organizer]

External Data: Agillic email guide (with Structure)	### TASK DESCRIPTION ### Write an email to the audience of a conference hosted last week. Follow the best practice bellow to improve your writting. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. IMPORTANT: You will be provided wlth a guide on email writing, closely follow its advice to improve your writting! ### TASK REQUIRED FORMAT ### The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass ### EMAIL GUIDE ### "Preheader text Highlight the most important content of your email or the content to catch the interest of your reader. This could be the top story of your newsletter or the reader's preferred product. We suggest aiming for a preheader text length of between 40-70 characters, including spaces, ***Full text shortned for brevity***	Subject: Recap of Prompt Engineering Masterclass and Access to Useful Materials Preview Text: Thank you for attending the conference! Access the sources and materials mentioned during the event. Dear Attendees of the Prompt Engineering Masterclass, We hope this email finds you well. We want to extend our sincerest gratitude to all of you who attended the Prompt Engineering Masterclass hosted at VŠE University in Prague last week. Your participation and engagement made the event a great success. As a token of appreciation, we would like to provide you with access to an article that includes the sources and useful materials mentioned during the conference. You can find the article by following this link: https://www.omnicrane.com/en/prompting-masterclass. We believe that the information shared during the conference will continue to be valuable to you in your professional endeavors. We encourage you to take the time to review the article and explore the resources provided. We are confident that you will find them to be beneficial and insightful. Once again, we want to express our gratitude for your attendance and contribution to the Prompt Engineering Masterclass. We hope to have the opportunity to connect with you again in the future. Best regards, [Your Name] [Your Position] [Conference Host Organization]

External Data: Summary of Best Practice by Claude Opus Model + Prompt Structure	### TASK DESCRIPTION ### Write an email to the audience of a conference hosted last week. Follow the best practice bellow to improve your writting. The goal is to recap the conference, thank the people who attended and encourage readers to open a link to an article including the sources and useful materials mentioned during the conference. The name of the conference is Prompt Engineering Masterclass, it was hosted at VŠE University in Prague. IMPORTANT: You will be provided with a set of best practice for writing. Write the email based on the insructions ### TASK REQUIRED FORMAT ### The format should include Subject, Preview Text and Text of email LINK TO ARTICLE: https://www.omnicrane.com/en/prompting-masterclass ### USE THE FOLLOWING BEST PRACTICE ABOUT WRITTING EMAILS ### Craft a compelling subject line: Keep it concise (under 50 characters) Use action-oriented language Personalize when possible Avoid spam trigger words Create a sense of urgency or curiosity Write a strong opening line: Grab the reader's attention immediately Personalize the greeting Clearly state the purpose of the email Use a conversational and friendly tone: Write as if you're speaking to the reader Use simple, easy-to-understand language Avoid jargon and complex terminology Structure your content for readability: Use short paragraphs (2-3 sentences max) Break up text with subheadings and bullet points Highlight key information with bold or italics Personalize the content: Segment your email list based on subscriber preferences Use the recipient's name when appropriate Tailor the message to the reader's interests and needs Focus on benefits, not just features: Explain how your product or service solves the reader's problems Highlight the value and benefits for the reader Use customer testimonials or case studies to build trust Include a clear and compelling call-to-action (CTA): Make the CTA button or link prominent and easy to find Use action-oriented language (e.g., "Get Started," "Download Now") Create a sense of urgency (e.g., "Limited	**Subject: Recap of Prompt Engineering Masterclass - Check out the Exclusive Article!** **Preview Text: Thank you for attending the conference last week. Click here to access the exclusive article with all the sources and materials mentioned during the event.** **Dear [Name],** **Thank you for joining us at the Prompt Engineering Masterclass hosted at VŠE University in Prague last week. It was a pleasure to have you as part of our audience.** **We are excited to share with you an exclusive article that includes all the sources and useful materials mentioned during the conference. You can access the article by clicking on the following link: https://www.omnicrane.com/en/prompting-masterclass** **In the article, you will find a comprehensive recap of the event, along with valuable insights and resources that will further enhance your understanding of the topics covered. We encourage you to take advantage of this opportunity to dive deeper into the content presented at the conference.** **We hope you find the article informative and beneficial to your professional development. If you have any questions or feedback, please feel free to reach out to us.** **Thank you once again for your participation, and we look forward to staying connected with you.** **Best regards,** **[Your Name]** **[Your Position]** **[Your Contact Information]**

	Time Offer") Ensure the CTA aligns with the email's purpose	

The end.